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Evaluating Value at Risk Methodologies: Accuracy versus Computational Time

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Abstract

Recent research has shown that different methods of computing Value at Risk (VAR) generate widely varying results, suggesting the choice of VAR method is very important. This article examines six VAR methods, and compares their computational time requirements and their accuracy when the sole source of inaccuracy is errors in approximating nonlinearity. Simulations using portfolios of foreign exchange options showed fairly wide variation in accuracy and unsurprisingly wide variation in computational time. When the computational time and accuracy of the methods were examined together, four methods were superior to the others. The article also presents a new method for using order statistics to create confidence intervals for the errors and errors as a per cent of true value at risk for each VAR method. This makes it possible to easily interpret the implications of VAR errors for the size of shortfalls or surpluses in a firm's risk-based capital.

1. Introduction

New methods of measuring and managing risk have evolved in parallel with the growth of the OTC derivatives market. One of these risk measures, known as Value at Risk (VAR) has become especially prominent, and now serves as the basis for the most recent BIS market-risk-based capital requirement. Value at Risk is usually defined as the largest loss in portfolio value that would be expected to occur due to changes in market prices over a given period of time in all but a small percentage of circumstances.[1] This percentage is referred to as the confidence level for the Value at Risk measure.[2] An alternative characterization of Value at Risk is the amount of capital the firm would require to absorb its portfolio losses in all but a small percentage of circumstances.[3]

Value at Risk's prominence has grown because of its conceptual simplicity and flexibility. It can be used to measure the risk of an individual instrument, or the risk of an entire portfolio. VAR measures are also potentially useful for the short-term management of the firm's risk. For example, if VAR figures are available on a timely basis, then a firm can increase its risk if VAR is too low, or decrease its risk if VAR is too high.

Although VAR is conceptually simple and flexible, in order for VAR figures to be useful, they also need to be reasonably accurate, and they need to be available on a timely enough basis so that they can be used for risk management. This will typically involve a tradeoff since the most accurate methods may take unacceptably long amounts of time to compute, while the methods that take the least amount of time are often the least accurate. The purpose of this article is to examine the tradeoffs between accuracy and computational

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time among six different methods of computing VAR, and to highlight the strengths and weaknesses of the various methods.

The need to examine accuracy is especially poignant, since the most recent Basle market risk capital proposal sets capital requirements based on VAR estimates from a firm's own internal VAR models. In addition it is important to examine accuracy since recent evidence on the accuracy of VAR methods has not been encouraging; different methods of computing VAR tend to generate widely varying results, suggesting that the choice of VAR method is very important. The differences in the results appear to be due to the use of procedures that implicitly use different methods to impute the joint distribution of the factor shocks (Hendricks, 1996; Beder, 1995),[4] and due to differences in the treatment of instruments whose payoffs are nonlinear functions of the factors (Marshall and Siegel, 1996).[5]

Tradeoffs between accuracy and computational time are most acute when portfolios contain large holdings of instruments whose payoffs are nonlinear in the underlying risk factors because VAR is the most difficult to compute for these portfolios. However, while the dispersion of VAR estimates for nonlinear portfolios has been documented, there has been relatively little publicly available research that compares the accuracy of the various methods. Moreover, the results that have been reported are often specific to a small number of portfolios, and the magnitude of errors is often reported only in dollar terms. Because errors reported in this way are not invariant to the scale of the portfolio, this reporting scheme makes it very difficult to compare results on the accuracy of the methods across the VAR literature, and also makes it difficult to relate the results on accuracy to the capital charges under the BIS market-risk-based capital proposal. While there has been little publicly available information on the accuracy of the methods, there has been even less reported on the computational feasibility of the various methods.

Distinguishing features of this article are its focus on accuracy versus computation time for portfolios that exclusively contain nonlinear instruments. An additional important contribution of this article is that it provides a statistically rigorous and easily interpretable method for gauging the accuracy of the VAR estimates in both dollar terms, and as a per cent of true VAR even though true VAR is not known.

Before proceeding, it is useful to summarize our empirical results. We examined accuracy and computational time in detail using simulations on test portfolios of foreign exchange options. The results on accuracy varied widely. One method (delta-gamma-minimization) produced results which consistently overstated VAR by very substantial amounts. The results on accuracy for the other methods can be roughly split into two groups: the accuracy of simple methods (delta, and delta-gamma-delta), and the accuracy of relatively complex methods (delta-gamma Monte Carlo, modified grid Monte Carlo, and Monte Carlo with full repricing). The simple methods were generally less accurate than the complex methods and the differences in the performance of the methods were substantial. The results on computational time favored the simple methods, which is not surprising. However, one of the complex methods (delta-gamma Monte Carlo) also takes a relatively short time to compute, and produces among the most accurate results. Based on the tradeoffs between accuracy and computational time, that method appears to perform the best for the portfolios considered here. However, even for this method, we were able to statistically detect a tendency to overstate or understate true VAR for about 25% of the 500 randomly chosen portfolios from simulation exercise 2. When VAR was over- or understated, a conservative estimate of the average magnitude of the error was about 10% of true VAR with a standard deviation of about the same amount. In addition, we found from simulation 1, that all VAR methods except for Monte Carlo with full repricing generated large errors as a per cent of VAR for deep out-of-the-money options with a short time to expiration. Although VAR for these options is typically very small, if portfolios do contain a large concentration of these options, then the best way to capture the risk of these options may be to use Monte Carlo with full repricing.[6]

The article proceeds in six sections. Section 2 formally defines Value at Risk and provides an overview of various methodologies for its computation. Section 3 discusses how to measure the accuracy of VAR computations. Section 4 discusses computational time requirements. Section 5 presents results from our simulations for portfolios of foreign exchange options. Section 6 concludes.

2. Methods for measuring Value at Risk

It is useful to begin our analysis of Value at Risk with its formal definition. Throughout, we will follow current practice and measure Value at Risk over a fixed span of time (normalized to one period) in which positions are assumed to remain fixed. Many firms set this span of time to be one day. This amount of time is long enough for risk over this horizon to be worth measuring, and it is short enough that the fixed position may reasonably approximate the firm's one-day or overnight position for risk management purposes.

To formally define Value at Risk requires some notation. Denote $V(P_t, X_t, t)$ as the value of portfolio V at time t with instruments X_t and instrument prices P_t and denote $\Delta V(P_{t+1} - P_t, X_t, t)$ as the change in portfolio value between period t and $t+1$. The cumulative density function of $\Delta V(P_{t+1} - P_t, X_t, t)$ conditional on X_t and time t information I_t is

$$G(k, I_t, X_t) = \text{Probability}(\Delta V(P_{t+1} - P_t, X_t) \leq k | I_t).$$

This allows the inverse cumulative density function for $\Delta V(P_{t+1} - P_t, X_t)$ to be defined as:

$$G^{-1}(u, I_t, X_t) = \inf\{k : G(k, I_t, X_t) = u\}.$$

Value at Risk for confidence level u is defined in terms of the inverse cumulative density function of ΔV:

$$VAR(u, I_t, X_t) = G^{-1}(u, I_t, X_t).$$

In words, Value at Risk at confidence level u is the largest loss that is expected to occur except for a set of circumstances with probability u. This is equivalent to defining Value at Risk at confidence level u as the uth quantile of the distribution of $\Delta V(P_{t+1} - P_t, X_t)$ given I_t. The definition of VAR highlights its dependence on the function G^{-1} which is a conditional function of the instruments X_t and the information set I_t.[7]

Value at Risk methods attempt to implicitly or explicitly make inferences about

$G^{-1}(u, I_t, X_t)$ in a neighborhood near confidence level u. In a large portfolio G^{-1} depends on the joint distribution of potentially tens of thousands of different instruments. This makes it necessary to make simplifying assumptions in order to compute VAR; these assumptions usually take three forms. First, the dimension of the problem is reduced by assuming that the price of the instruments depends on a vector of factors f that are the primary determinants of changes in portfolio value.[8] This allows changes in portfolio value to be expressed as $\Delta V(\varepsilon_{t+1}, X_t, t)$ where $\varepsilon_{t+1} = f_{t+1} - f_t$. Second, $\Delta V(\varepsilon_{t+1}, X_t, t)$ is usually approximated instead of being calculated explicitly. Finally, convenient functional forms are often assumed for the distribution of ε_{t+1}.

Each of these simplifying assumptions is likely to introduce errors into the VAR estimates. The first assumption induces errors if an incorrect or incomplete set of factors is chosen; the second assumption introduces approximation error; and the third assumption introduces error if the wrong distribution of ε_{t+1} is chosen. The first and third sources of error are model error while the errors of the second type are approximation error. Because model error can take so many forms, we choose to abstract from it here by assuming that the factors have been chosen correctly, and the distribution of the factor innovations is correct. This allows us to focus on the errors induced by the approximations used in each VAR method, and to get a clean measure of these errors. In future work, I hope to incorporate model error in the analysis and examine whether some methods of computing VAR are robust to certain classes of model errors.

The different methods of computing VAR are distinguished by their simplifying assumptions. The Value at Risk methodologies that I consider fall into two broad categories. The first are delta and delta-gamma methods. These methods typically make assumptions that lead to analytic or near analytic tractability for the Value at Risk computation. The second set of methods use Monte Carlo simulation to compute Value at Risk. Their advantage is that they are capable of producing very accurate estimates of Value at Risk; however, these methods are not analytically tractable, and they are very time and computer intensive.

2.1. Delta and delta-gamma methods

The delta and delta-gamma methods typically make the distributional assumption that changes in the factors are distributed normally conditional on today's information:

$$(\varepsilon_{t+1} | I_t) \sim N(0, \Sigma).$$

The delta and delta-gamma methods principally differ in their approximation of change in portfolio value. The delta method approximates the change in portfolio value using a first-order Taylor series expansion in the factor shocks and the time horizon over which VAR is computed:[9]

$$\Delta V(\varepsilon_{t+1}, X_t) \approx \delta^{\top} \varepsilon_{t+1} + \theta.$$

$$\text{where } \delta = \left.\frac{\partial V(\varepsilon_{t+1,}X_t, t)}{\partial \varepsilon_{t+1}}\right|_{\{\varepsilon_{t+1}=0\}}$$

$$\theta = \left.\frac{\partial V\ (\varepsilon_{t+1}, X_t, t)}{\partial t}\right|_{\{\varepsilon_{t+1}=0\}}$$

Delta-gamma methods approximate changes in portfolio value using a Taylor series expansion that is second order in the factor shocks and first order in the VAR time horizon:

$$\Delta V(\varepsilon_{t+1}, X_t) \approx \delta^\top \varepsilon_{t+1} + .5\varepsilon_{t+1}^\top \Gamma \varepsilon_{t+1} + \theta$$

$$\text{where } \Gamma = \left.\frac{\partial^2 V(\varepsilon_{t+1}, X_t)}{\partial \varepsilon_{t+1} \partial \varepsilon_{t+1}{}^\top}\right|_{\{\varepsilon_{t+1}=0\}}$$

At this juncture it is important to emphasize that the Taylor expansions and distributional assumptions depend on the choice of factors. This choice can have an important impact on the ability of the Taylor expansions to approximate nonlinearity,[10] and may also have an important impact on the adequacy of the distributional assumptions.

For a given choice of the factors, the first-order approximation and distributional assumption used in the *delta* method[11] imply $\Delta V(\varepsilon_{t+1}, X_t) \sim N(\theta, \delta'\Sigma\delta)$. Therefore calculating VAR using this method requires only calculating the inverse cumulative density function of the normal distribution. In this case, elementary calculation shows Value at Risk at confidence level u is $\theta + \Phi^{-1}(u)\sqrt{\delta\Sigma\delta}$, where Φ^{-1} is the inverse of the cumulative density function of the normal distribution.

The delta method's main virtue is its simplicity. This allows Value at Risk to be computed very rapidly. However, its linear Taylor series approximation may be inappropriate for portfolios whose value is a nonlinear function of the factors. The normality assumption is also suspect since many financial time series are fat-tailed.[12] The delta-gamma approaches attempt to improve on the delta method by using a second-order Taylor series to capture the nonlinearities of changes in portfolio value. Some of the delta-gamma methods also allow for more flexible assumptions on the distribution of the factor innovations.

Delta-gamma methods represent some improvement over the delta method by offering a more flexible functional form for capturing nonlinearity.[13] However, a second-order approximation is still not capable of capturing all of the nonlinearities of changes in portfolio value. In addition, the nonlinearities that are captured come at the cost of reduced analytic tractability relative to the delta method. The reason is that changes in portfolio value using the second-order Taylor series expansion are not approximately normally distributed even if the factor innovations are.[14] This loss of normality makes it more difficult to compute the G^{-1} function than in the delta method. Two basic fixes are used to try to finesse this difficulty. The first abandons analytic tractability and approximates G^{-1} using Monte Carlo methods; by contrast, the second attempts to maintain analytic or near-analytic tractability by employing additional distributional assumptions.

Many of the studies that use a delta-gamma method for computing VAR refer to the method they use as *the* delta-gamma method. This nomenclature is inadequate here since I examine the performance of three delta-gamma methods in detail, as well as discussing two recent additions to these methods. Instead, I have chosen names that are somewhat descriptive of how these methods are implemented.

The first approach is the *delta-gamma Monte Carlo* method.[15] Under this approach, a large number of realizations of ε_{t+1} are drawn from its distribution[16] and for each realization $\Delta V(\varepsilon_{t+1}, I_t, X_t)$ is computed according to the second-order delta-gamma Taylor series approximation. The empirical cumulative density function of the resulting series of ΔVs is an approximation of the cumulative density function G. The value of $\Delta V(\varepsilon_{t+1}, I_t, X_t)$ corresponding to the uth percentile of the empirical distribution is the estimate of Value at Risk. If the second-order Taylor approximation is exact, then this estimator will converge to true Value at Risk as the number of Monte Carlo draws approaches infinity.

The next approach is the *delta-gamma-delta* method. I chose to give it this name because it is a delta-gamma based method that maintains most of the simplicity of the delta method. It does so by making the distributional assumption that ε_{t+1} aud the unique elements of $\varepsilon_{t+1}\varepsilon_{t+1}^{\mathrm{T}}$ are uncorrelated and normally distributed shocks to portfolio value with portfolio sensitivities corresponding to the elements of δ and Γ that they multiply in the second-order Taylor series expansion.[17] For example, if V is a function of one factor, then the shocks to V are ε_{t+1} and ε_{t+1}^2, and their assumed distribution is:

$$\begin{bmatrix} \varepsilon_{t+1} \\ \varepsilon_{t+1}^2 \end{bmatrix} \sim N\left(\begin{bmatrix} 0 \\ \sigma^2 \end{bmatrix}, \begin{bmatrix} \sigma^2 & 0 \\ 0 & 2\sigma^4 \end{bmatrix}\right)$$

The mean and covariance matrix in the above expressions are correct if $\varepsilon_{t+1} \sim N(0, \sigma^2)$, but the assumption of normality cannot possibly be correct,[18] and should instead be viewed as a convenient assumption that simplifies the computation of VAR. Under this distributional assumption $\Delta V(\varepsilon_{t+1}, X_t) \sim N(\theta + 0.5\Gamma\sigma^2, \delta^2\sigma^2 + 0.5\Gamma^2\sigma^4)$.[19] Analogous to the delta method, G^{-1} and hence VAR is simple to calculate in the delta-gamma-delta approach. In this example VAR is equal to $\theta + 0.5\Gamma\sigma^2 + \Phi^{-1}(u)\sqrt{\delta^2\sigma^2 + 0.5\Gamma^2\sigma^4}$. The calculation of Value at Risk for a large number of factors proceeds along similar lines. However, when there are N true factors, and no elements in the Γ matrix are restricted to be zero, the number of factors that are used in the delta-gamma-delta approach is $N^2 + 3N/2$. For moderate size N, this becomes unwieldy. When I implemented this approach, I used the transformation in Wilson (94) to express ΔV as $\delta^{*\top}u^* + 0.5u^{*\top}Du^*$ where the vector $u^* \sim N(0, \mathrm{I})$ and D is diagonal. Using this transformation reduces the number of effective factors to $2N$.

The next delta-gamma approach is the *delta-gamma-minimization* method discussed in Wilson (1994). It maintains the assumptions that ε_{t+1} is normally distributed, and that changes in portfolio value are well-approximated quadratically. It then finds the particular value of ε_{t+1} that creates the greatest loss in portfolio value subject to the constraint that ε_{t+1} lies within a sphere centered at 0 containing $1 - u\%$ of the probability mass of the distribution of ε_{t+1}. In mathematical terms, this Value at Risk estimate is the solution to the minimization problem:

$$VAR = \min_{\varepsilon_{t+1}} \delta^\top \varepsilon_{t+1} + 0.5\varepsilon_{t+1}^\top \Gamma \varepsilon_{t+1}$$

$$\text{such that } \varepsilon_{t+1}^\top \textstyle\sum^{-1} \varepsilon_{t+1} \leq c * (1 - u, k)$$

where $c^*(1-u,k)$ is the $u\%$ critical value of the central chi-squared distribution with k degrees of freedom; i.e. $1-u\%$ of the probability mass of the central chi-squared distribution with k degrees of freedom is below $c^*(1-u,k)$.

This approach makes a very strong but not obvious distributional assumption to identify Value at Risk. The distributional assumption is that the value of the portfolio outside of the constraint set is lower than anywhere within the constraint set. If this assumption is correct, and the quadratic approximation for changes in portfolio value is exact, then the Value at Risk estimate would exactly correspond to the uth quantile of the change in portfolio value. Unfortunately there is little reason to believe that this key distributional assumption will be satisfied, and it is realistic to believe that this condition will be violated for a substantial (i.e., nonzero) proportion of the shocks that lie outside the constraint set.[20] This means that typically less than $u\%$ of the realizations are below the estimated Value at Risk, and thus the delta-gamma minimization method is very likely to overstate true Value at Risk. In extreme cases the overstatement could be very large. For example, suppose ΔV has the same cumulative density function as a $N(0,1)$ variable, but that the lowest value of ΔV occurs for a factor shock that lies within the 95% constraint set. In this case, estimated Value at Risk would be minus infinity at the 5% confidence level, while true Value at Risk would be -1.645. This example is extreme, but illustrates the point that this method is capable of generating large errors. It is more likely that at low levels of confidence, as the amount of probability mass outside the constraint set goes to zero, the estimate of Value at Risk using this method remains conservative but converges to true Value at Risk (conditional on the second-order approximation being exact). This suggests the delta-gamma-minimization method will produce better (i.e., less conservative) estimates of Value at Risk when the confidence level u is small.

The advantage of the delta-gamma-minimization approach is that it does not rely on the incorrect joint normality assumption that is used in the delta-gamma-delta method, and does not require the large number of Monte Carlo draws required in the delta-gamma Monte Carlo approach. Whether these advantages offset the errors induced by this approach's conservatism is an empirical question.

In the sections that follow I will examine the accuracy and computational time requirements of the delta-gamma Monte Carlo, delta-gamma-delta, and delta-gamma-minimization methods. For completeness I mention here two additional delta-gamma methods. The first of these was recently introduced by Peter Zangari of J. P. Morgan (1996) and parameterizes a known distribution function so that its first four moments match the first four moments of the delta-gamma approximation to changes in portfolio value. The cumulative density function of the known distribution function is then used to compute Value at Risk. I will refer to this method as the *delta-gamma-Johnson* method because the statistician Norman Johnson introduced the distribution function that is used. The second approach calculates VAR by approximating the uth quantile of the distribution of the delta-gamma approximation using a Cornish–Fisher expansion. This approach is used in Fallon (1996) and Zangari (1996). This approach is similar to the first except that a Cornish–Fisher expansion approximates the quantiles of an unknown distribution as a function of the quantiles of a known distribution function and the cumulants of the known and unknown distribution functions.[21] I will refer to this approach as the *delta-gamma-Cornish–Fisher* method.

Both of these approaches have the advantage of being analytic. The key distributional assumption of both approaches is that matching a finite number of moments or cumulants of a known distribution with those of an unknown distribution will provide adequate estimates of the quantiles of the unknown distribution. Both of these approaches are likely to be less accurate than the delta-gamma Monte Carlo approach (with a large number of draws) since the Monte Carlo approach implicitly uses all of the information on the cumulative distribution function of the delta-gamma approximation while the other two approaches throw some of this information away. In addition, as we will see below, the delta-gamma Monte Carlo method (and other Monte Carlo methods) can be used with historical realizations of the factor shocks to compute VAR estimates that are not tied to the assumption of conditionally normally distributed factor shocks. It would be very difficult to relax the normality assumption for the delta-gamma-Johnson and delta-gamma-Cornish–Fisher methods because without these assumptions the moments and cumulants of the second-order Taylor series for ΔV would be unknown and very difficult to compute.

2.2. *Monte Carlo methods*

The next set of approaches for calculating Value at Risk are based on Monte Carlo simulation. Monte Carlo simulation works by using a series of random draws of the factor shocks (ε_{t+1}). These shocks combined with some approximation method are used to generate a random series of changes in portfolio value (ΔV). The empirical cumulative density function of the changes in portfolio value, $\hat{G}$, is then used as a proxy for G, and $\hat{G}^{-1}(u)$ is the corresponding estimate of VAR at confidence level u.

The Monte Carlo approaches have the potential to improve on the delta and delta-gamma methods because they can allow for alternative methods of approximating changes in portfolio value, and they can allow ε_{t+1} to have a nonnormal distribution. The Monte Carlo methodologies considered here allow for two methods for approximating change in portfolio value and two methods of parameterizing the distribution of ε_{t+1}. These approaches can be combined.

The first approximation method is the *full Monte Carlo* method.[22] This method uses exact pricing for each Monte Carlo draw, thus eliminating errors from approximations to ΔV. Since this method involves no approximation error, $\hat{G}$, the full Monte Carlo estimate of G converges to G in probability as the sample size grows provided the distributional assumptions are correct. Consequently, $\hat{G}^{-1}(u)$ converges in probability to true Value at Risk at confidence level u. Because this approach produces good estimates of VAR for large sample sizes, the full Monte Carlo estimates are a good baseline against which to compare other methods of computing Value at Risk. The downside of the full Monte Carlo approach is that it tends to be very time-consuming, especially if analytic solutions for some assets prices don't exist.

The second approximation method is the *grid Monte Carlo* approach.[23] In this method a grid of realizations for ε_{t+1} (for N factors, this would involve an N-dimensional grid) is created and the change in portfolio value is calculated exactly for each node of the grid. To make approximations using the grid, for each Monte Carlo draw, the factor shocks should lie somewhere on the grid, and changes in portfolio value for these shocks can be estimated by interpolating from changes in portfolio value at nearby nodes. This interpolation method is likely to provide a better approximation than either the delta or delta-gamma methods because it places fewer restrictions on the behavior of the nonlinearity and because it becomes ever less restrictive as the mesh of the grid is increased.

As I will discuss in the next section, grid Monte Carlo methods suffer from a curse of dimensionality problem since the number of grid points grows exponentially with the number of factors. To avoid the dimensionality problem, I model the change in the value of an instrument by combining a one-dimensional grid to model the behavior of the factor shock that has the most nonlinear effect on instrument value with a first-order Taylor series to capture the effect of the other factors.[24]

There are two basic approaches that can be used to make the Monte Carlo draws. The first approach is to draw realizations of ε_{t+1} from some distribution that is assumed to be the true distribution for ε_{t+1}. This approach relaxes the normality assumption but is likely to introduce some error because the true distribution of ε_{t+1} is probably unknown. The second approach is to make the Monte Carlo draws via a historical simulation approach (this is also known as bootstrapping). In this approach, draws of ε_{t+1} are simulated from historical realizations of ε that are believed to be draws from the same conditional distribution as ε_{t+1}. This will require that past realization of ε are observable, and it will probably require observability of the past factor realizations.[25] It will also be necessary to make some assumptions that link the stochastic process that generated draws of ε historically to the process that generates the draws for time period $t+1$.[26] The main advantage of this approach is that even when the true distribution of ε_{t+1} is not known, it may still be possible to make draws from it. The disadvantage of this approach is that there may not be enough historical data to make the draws, or there is no past time period that is sufficiently similar to the present for the historical draws to provide information about the future distribution of ε_{t+1}.

The historical simulation approach will not be compared with other Monte Carlo approaches in this article. I plan to pursue this topic in another paper. Here, we will examine the grid Monte Carlo approach and the full Monte Carlo approach for a given distribution for ε_{t+1}.

3. Measuring the accuracy of VAR estimates

Estimates of VAR are generally not equal to true VAR, and thus should ideally be accompanied by some measure of estimator quality such as statistical confidence intervals or a standard error estimate. This is typically not done for delta and delta-gamma based estimates of VAR since there is no natural method for computing a standard error or constructing a confidence interval.

In Monte Carlo estimation of VAR, anecdotal evidence suggests that error bounds are typically not provided, but that error is controlled somewhat by making Monte Carlo draws until the VAR estimates do not change significantly in response to additional draws. This procedure may be inappropriate for VAR calculation since Value at Risk is likely to depend on extreme draws which are made infrequently. Put differently, Value at Risk estimates may change little in response to additional Monte Carlo draws even if the Value at Risk estimates are poor.

Although error bounds are typically not provided for full Monte Carlo estimates of VAR, it is not difficult to use the empirical distribution from a sample size of N Monte Carlo draws to form confidence intervals for Monte Carlo Value at Risk estimates (we will form 95% confidence intervals). Given the width of the interval, it can be determined whether the sample of Monte Carlo draws should be increased to provide better estimates of VAR. The confidence intervals that we will discuss have the desirable properties of being nonparametric (i.e., they are valid for any continuous distribution function G), based on finite sample theory, and simple to compute. There is one important requirement: the draws from the distribution of ε_{t+1} must be independently and identically distributed (i.i.d.).

The confidence intervals for the Monte Carlo estimates of Value at Risk can also be used to construct confidence intervals for the error and percentage error from computing VAR by other methods. These confidence intervals are extremely useful for evaluating the accuracy of both Monte Carlo methods and any other method of computing VAR. The use of confidence intervals should also improve on current practices of measuring VAR error. The error from a VAR estimate is often calculated as the difference between the estimate and a Monte Carlo estimate. This difference is an incomplete characterization of the VAR error, since the Monte Carlo estimate is itself measured imperfectly. The confidence intervals give a more complete picture of the VAR error because they take the errors from the Monte Carlo into account.[27]

Table 1 provides information on how to construct 95% confidence intervals for Monte Carlo estimates of VAR for VAR confidence levels of 1 and 5%. To illustrate the use of the table, suppose one makes 100 i.i.d. Monte Carlo draws and wants to construct a 95% confidence interval for Value at Risk at the 5% confidence level (my apologies for using confidence two different ways). Then, the upper right column of table 1 shows that the largest portfolio loss from the Monte Carlo simulations and the 10th largest portfolio loss form a 95% confidence interval for true Value at Risk. The parentheses below these figures restate the confidence bounds in terms of percentiles of the Monte Carlo distribution. Hence, the first percentile and tenth percentile of the Monte Carlo loss distribution bound the fifth percentile of the true loss distribution with 95% confidence when 100 draws are made. If the spread between the largest and 10th largest portfolio loss are too high, then a tighter confidence interval is needed, which will require more Monte Carlo draws. Table 1 provides upper and lower bounds for larger numbers of draws and for VAR confidence levels of 1% and 5%. As one would expect, the table shows that as the number of draws increases, the bounds, measured in terms of percentiles of the Monte Carlo distribution, tighten, so that with 10,000 draws, the 4.57th and 5.44th percentile of the Monte Carlo distribution bound the 5th percentile of the true distribution with 95% confidence.

Table 1 can also be used to construct confidence intervals for the error and percentage errors from computing VAR using Monte Carlo or other methods. Details on how the entries in table 1 were generated and details on how to construct confidence intervals for percentage errors are contained in Appendix 1.

Table 1. Nonparametric 95% confidence intervals for Monte Carlo VAR figures.

Number of draws	VAR Confidence Level 1%		VAR Confidence Level 5%	
	Lower bound	Upper bound	Lower bound	Upper bound
100	—	—	1	10
	—	—	(1%)	(10.%)
300	1	11	8	23
	(0.33%)	(3.67%)	(2.7%)	(7.7%)
500	1	10	15	35
	(0.2%)	(2.0%)	(3.0%)	(7.0%)
1,000	4	17	37	64
	(0.4%)	(1.7%)	(3.7%)	(6.4%)
10,000	81	120	457	544
	(0.81%)	(1.2%)	(4.57%)	(5.44%)
50,000	456	545	2404	2597
	(0.912%)	(1.09%)	(4.81%)	(5.19%)
100,000	938	1063	4865	5136
	(0.938%)	(1.063%)	(4.865%)	(5.136%)
250,000	2402	2599	12,286	12,715
	(0.9608%)	(1.0396%)	(4.9144%)	(5.086%)
500,000	4862	5139	24,698	25,303
	(0.9742%)	(1.0278%)	(4.9396%)	(5.0606%)
1,000,000	9805	10196	49,573	50,428
	(0.9805%)	(1.0196%)	(4.9573%)	(5.0428%)

Notes. Colums (2)–(3) and (4)–(5) report the index of order statistics that bound the first and fifth percentile of an unknown distribution with 95% confidence when the unknown distribution is simulated using the number of i.i.d. Monte Carlo draws indicated in column (1). If the unknown distribution is for changes in portfolio value, then the bounds form a 95% confidence interval for Value at Risk at the 1% and 5% level. The figures in parentheses are the percentiles of the Monte Carlo distribution that correspond to the order statistics. For example, the figure in colums (4)–(5) of row (1) show that the first and tenth order statistic from 100 i.i.d. Monte Carlo draws bound the 5th percentile of the unknown distribution with 95% confidence. Or, as shown in parentheses, with 100 i.i.d. Monte Carlo draws, the 1st and 10th percentile of the Monte Carlo distribution bound the fifth percentile of the true distribution with 95% confidence. No figures are reported in colums (2)–(3) of the first row because it is not possible to create a 95% confidence interval for the first percentile with only 100 observations.

4. Computational considerations

Risk managers often indicate that complexities of various VAR methods limit their usefulness for daily VAR computation. The purpose of this section is to investigate and provide a very rough framework for categorizing the complexity of various VAR calculations. As a first pass at modeling complexity, I segregate Value at Risk computations into two basic types. The first type is the computations involved in approximating the value of the portfolio and its derivatives, while the second is the computations involved in estimating the parameters of the distribution function of ε_{t+1}. We will not consider the second type here since this information should already be available when Value at Risk is to be computed. We will consider the first type of computation. This first type can be further segregated into two groups. The first are *complex* computations that involve the pricing of an instrument or the computation of a partial derivative such as δ or Γ. The second are *simple* computations; these involve relatively simple computations such as linear interpolation or using the inputs from complex computations to compute VAR.[28] For the purposes of this article I have assumed that each time the portfolio is repriced using simple techniques, such as linear interpolation, it involves a single simple computation. I also assume that once delta and gamma for a portfolio have been calculated, computing VAR from this information requires one simple computation. This is a strong assumption, but it is probably not a very important assumption because the bulk of the computational time is not driven by the simple computations, but by the complex computations.

I also make assumptions about the number of complex computations that are required to price an instrument. In particular, I will assume that the pricing of an instrument requires a single complex computation no matter how the instrument is actually priced. This assumption is clearly inappropriate for modeling computational time in some circumstances,[29] but I make this assumption here to abstract from the details of the actual pricing of each instrument.

The amount of time required to calculate VAR depends on the number and complexity of the instruments in the portfolio, the method used to calculate VAR, and the amount of parallelism in the firm's computing structure. To illustrate these points in a single unifying framework, let V denote a portfolio whose value is sensitive to N factors, and which contains a total of I different instruments. An instrument's complexity depends on whether its price has a closed-form solution, and on the number of factors used to price the instrument. For purposes of simplicity, I will assume that δ and Γ have closed-form solutions, for all instruments or no instruments, and present results for both of these extreme cases. To model the other dimension of complexity, let I_n denote the number of *different* instruments whose value is sensitive to n factors.[30] It follows that the number of instruments in the portfolio is $I = \sum_{n=1}^{N} I_n$.

To calculate δ and Γ for the portfolio, it is necessary to calculate δ and Γ for individual instruments and then combine the results. I assume the number of computations required to calculate δ analytically for an instrument with n factors is n, and the number of numerical computations required is $2n$. Similarly, to calculate Γ for an instrument with n factors analytically requires $n(n+1)/2$ computations and the number of numerical calculations is $3n + 2n(n-1)$.[31,32] This implies that the number of complex computations required for the delta approach is:

$$\#\ \text{Delta} = \begin{cases} \sum_{n=1}^{N} I_n n & \text{if analytical calculations} \\ \sum_{n=1}^{N} I_n 2n & \text{if numerical calculations} \end{cases}.$$

Similarly, the number of complex computations required for all of the delta-gamma approaches is equal to:[33]

$$\#\ \text{Delta} - \text{Gamma} = \begin{cases} \sum_{n=1}^{N} I_n (n^2+3n)/2 & \text{if analytical calculations} \\ \sum_{n=1}^{N} I_n (2n^2+3n) & \text{if numerical calculations} \end{cases}.$$

The grid Monte Carlo approach prices the portfolio using exact valuation at a grid of factor values and prices the portfolio at points between grid values using some interpolation technique (such as linear). For simplicity, I will assume that the grid is computed for k realizations of each factor. This then implies that an instrument that is sensitive to n factors will need to be repriced at k^n grid points. This implies that the number of complex computations that is required is:

$$\#\text{ Grid Monte Carlo} = \sum_{n=1}^{N} I_n k^n.$$

The number of computations required for the grid Monte Carlo approach is growing exponentially in n. For even moderate sizes of n, the number of computations required to compute VAR using grid Monte Carlo is very large. Therefore, it will generally be desirable to modify the grid Monte Carlo approach. I do so by using a method that I will refer to as the modified grid Monte Carlo approach. The derivation of this method is contained in footnotes of the discussion of the grid Monte Carlo method. This involves approximating the change in instrument value as the sum of the change in instrument value due to the changes in a set of factors that are allowed to enter nonlinearly plus the sum of the change in instrument value due to other factors that are allowed to enter linearly. For example, if m factors are allowed to enter nonlinearly, and each factor is evaluated at k values, then the change in instrument value due to the ''nonlinear'' factors is approximated on a grid of points using linear interpolation on k^m grid points while all other factors are held fixed. The change in instrument value due to the other n–m factors is approximated using a first-order Taylor series while holding the ''nonlinear'' instruments fixed.

The number of complex computations required for one instrument using the modified grid Monte Carlo approach is $k^m + n - m$ if the first derivatives in the Taylor series are computed analytically, and $k^m + 2(n - m)$ if they are computed numerically. Let $I_{m,n}$ denote the number of instruments with n factors that have m factors that enter nonlinearly in the modified grid Monte Carlo approach. Then, the number of complex computations required in the modified grid Monte Carlo approach with a grid consisting of k realizations per factor on the grid is:

$$\#\text{ Modified Grid Monte Carlo} = \begin{cases} \sum_{n=1}^{N} \sum_{m=1}^{N} I_{m,n}[k^m + n - m] & \text{if analytical calculations} \\ \sum_{n=1}^{N} \sum_{m=1}^{n} I_{m,n}[k^m + 2(n - m)] & \text{if numerical calculations} \end{cases}$$

Finally, the number of computations required for the exact pricing Monte Carlo approach (labeled Full Monte Carlo) is the number of draws made times the number of instruments that are repriced:

$$\#\text{ Full Monte Carlo} = NDRAWS \sum_{n=1}^{N} I_n.$$

Only one simple computation is required in the delta approach and in most of the delta-gamma approaches. The number of simple computations in the grid Monte Carlo approach is equal to the expected number of linear interpolations that are made. This should be equal to the number of Monte Carlo draws. Similarly, the delta-gamma Monte Carlo approach involves simple computations because it only involves evaluating a Taylor series with linear and quadratic terms.[34] The number of times this is computed is again equal to the number of times the change in portfolio value is approximated.

To obtain a better indication of the computational requirements for each approach,

Table 2. Numbers of complex and simple computations for VAR methods.

VAR method	# Complex computations	# Simple computations
Delta	6,000	1
DGD	23,000	1
DGMIN	23,000	1
DGMC	23,000	300
GridMC	1,110,000	300
ModGridMC	33,000	300
FullMC	900,000	0

Notes: The table illustrates the number of complex and simple computations required to compute Value at Risk using different methods for a portfolio with 3000 instruments. 1000 instruments are sensitive to 1 factor; 1000 instruments are sensitive to 2 factors; and 1000 instruments are sensitive to 3 factors. The distinction between simple and complex computations is described in the text. The methods for computing VAR are the Delta method (Delta), the Delta-Gamma-Delta method (DGD), the Delta-Gamma- Minimization method (DGMIN), the Delta-Gamma Monte Carlo method (DGMC), the Grid Monte Carlo method (GridMC), the Modified Grid Monte Carlo method (ModGridMC), and full Monte Carlo (FullMc).

consider a sample portfolio with 3000 instruments, where 1000 are sensitive to one factor, 1000 are sensitive to two factors, and 1000 are sensitive to three factors. We will implement the modified grid Monte Carlo approach with only one ''nonlinear'' factor per instrument. Both grid Monte Carlo approaches will create a grid with ten factor realizations per factor considered.[35] Finally, 300 Monte Carlo draws will be made to compute Value at Risk. The number of complex and simple computations required to compute VAR under these circumstances is provided in table 2.

The computations in the table are made for a small portfolio which can be valued analytically. Thus, the results are not representative of true portfolios, but they are indicative of some of the computational burdens imposed by the different techniques.[36] The most striking feature of the table is the large number of *complex* computations required in the grid and Monte Carlo approaches, versus the relatively small number in the delta and delta gamma approaches. The grid approach involves a large number of computations because the number of complex computations for each instrument is growing exponentially in the number of factors to which the instrument is sensitive. For example, an asset that is sensitive to one factor is priced 10 times, while an asset that is sensitive to three factors is priced 1000 times. Reducing the mesh of the grid will reduce computations significantly, but the number of computations is still substantial. For example, if the mesh of the grid is reduced by half, 155,000 complex computations are required for the grid approach. The modified grid Monte Carlo approach ameliorates the curse of dimensionality problem associated with grid Monte Carlo in a more satisfactory way by modeling fewer factors on a grid. This will definitely hurt the accuracy of the method, but it should very significantly improve on computational time when pricing instruments that are sensitive to a large number of factors.

The full Monte Carlo approach requires a large number of computations because each asset is priced 300 times using the Monte Carlo approach. However, the number of Monte Carlo calculations increases only linearly in the number of instruments and does not depend on the number of factors; therefore in large portfolios it may involve a smaller number of computations than some grid based approaches.

The delta and delta-gamma approaches require a much smaller number of calculations because the number of required complex computations is increasing only linearly in the number of factors per instrument for the delta method and only quadratically for the delta-gamma methods. As long as the number of factors per instrument remains moderate, the number of computations involved in the delta and delta-gamma methods should remain low relative to the other methods.

An important additional consideration is the amount of parallelism in the firm's process for computing value at risk. Table 2 shows that a large number of computations are required to calculate value at risk for all the methods examined here. If some of these computations are split among different trading desks throughout the firm and then individual risk reports are funneled upward for a firmwide calculation, then the amount of time required to compute Value at Risk could potentially be smaller than implied by the gross number of complex computations. For example, if there are 20 trading desks and each desk uses one hour to calculate its risk reports, it may take another hour to combine the results and generate a value at risk number for the firm, for a total of two hours. If instead, all the computations are done by the risk management group, it may require 21 hours to calculate Value at Risk, and the figures may be irrelevant to the firm's day-to-day operations once they become available.[37]

It is conceptually possible to perform all of the complex computations in parallel for all methods that have been discussed above. However, it will be somewhat more involved in the grid approaches and in the Monte Carlo approaches since individual desks need a priori knowledge of the grid points so they can value their books at those points. They may also need a priori knowledge of the Monte Carlo draws so that they can value their books at the draw points.[38]

To further evaluate the tradeoff between computational time and accuracy simulations are conducted in the next section.

5. Empirical analysis

Two simulation exercises were conducted to evaluate the accuracy of the VAR methods; a separate analysis examined computational time. All simulations computed VAR in units of U.S. dollars for portfolios of European foreign exchange options that were priced using the Black–Scholes model. Exchange rates and interest rates were treated as stochastic pricing variables; these were then mapped to RiskMetrics^{TM} risk factors and the covariance matrix of the risk factors was extracted from the RiskMetrics^{TM} regulatory data set. Implied volatility was treated as fixed. Estimates of VAR were computed using the methods described in section 2, and all of the partial- and second- and cross-partial derivatives used in the delta and delta-gamma methods were computed numerically.[39] All of the Monte Carlo methods used 10,000 Monte Carlo draws to compute Value at Risk.

5.1. Simulation exercises 1 and 2

Simulation exercise 1 examined VAR for positions in a single option that were either long or short a U.S. dollar/French franc ($/FFR) call or put that delivered 1 million French francs if exercised. For each option position, VAR was calculated for a set of seven evenly spaced moneynesses that ranged from 30% out of the money to 30% in the money, and for a set of 10 evenly spaced maturities that ranged from 0.1 years to 1.0 years.

Simulation exercise 2 examined VAR for 500 randomly chosen portfolios of foreign exchange rate options. The moneyness of each option was chosen as in simulation 1. The maturity of the options ranged from 18 days to 1 year. Each option's underlying exchange rate was chosen randomly from among the exchange rates between the currencies of Belgium, Canada, Switzerland, the United States, Germany, Spain, France, Great Britain, Italy, Japan, The Netherlands, and Sweden. The probability that the exchange rate between a currency pair was chosen was proportional to the amount of turnover in the OTC foreign exchange derivatives market in that currency pair relative to the other currency pairs considered. The measures of turnover are based on table 9G of the *Central Bank Survey of Foreign Exchange and Derivatives Market Activity 1995*. Additional details on the simulations are contained in Appendix 2.

5.2. The empirical accuracy of the methods

Results from simulation 1 for positions in the long call option for VAR at the 1% confidence level are presented in figures 1–5 and in table 3; results for other option positions will be selectively discussed, but not formally presented, in order to save space.[40]

Figure 1 presents plots of estimated VAR against moneyness and time to expiration for all six methods. The VAR estimates are qualitatively similar; for five methods they range from a low near 0 for deep out-of-the-money options to a high near $3500. However, the delta-gamma-minimization method generates substantially higher estimates, suggesting it grossly overstates VAR. This is consistent with its predicted bias towards overstating VAR.

Figures 2 and 3 plot the upper and lower bounds respectively of a 95% confidence interval for the error made using each VAR estimate. Figures 4 and 5 plot the upper and lower bounds of 95% confidence intervals for the percentage errors associated with each estimator.[41] These confidence intervals are derived in Appendix 1 and are useful for hypotheses testing. If a confidence interval contains 0, the null of no error cannot be rejected at a 5% significance level. If the interval is bounded above by 0 (i.e., if the upper bound of the confidence interval is below 0), the null hypothesis of no error is rejected in favor of the estimate understating true VAR, and if the interval is bounded below by zero (i.e., if the lower bound of the interval is above zero), the null of no error is rejected in favor of VAR being understated.

The upper bound confidence intervals in figures 2 and 4 are almost always above zero, showing that VAR is rarely understated for the long call-option positions. However, figures 3 and 5 show the delta and delta-gamma delta methods overstate VAR for at or near the money call options. This is not surprising. The value of a long, call-option declines at a decreasing rate in the underlying exchange rate. Because the delta method does not account for this effect, it would be expected to overstate VAR. This problem should be especially acute for at the money options where the partial derivative of option price with respect to the underlying exchange rate changes the most rapidly. The delta-gamma-delta

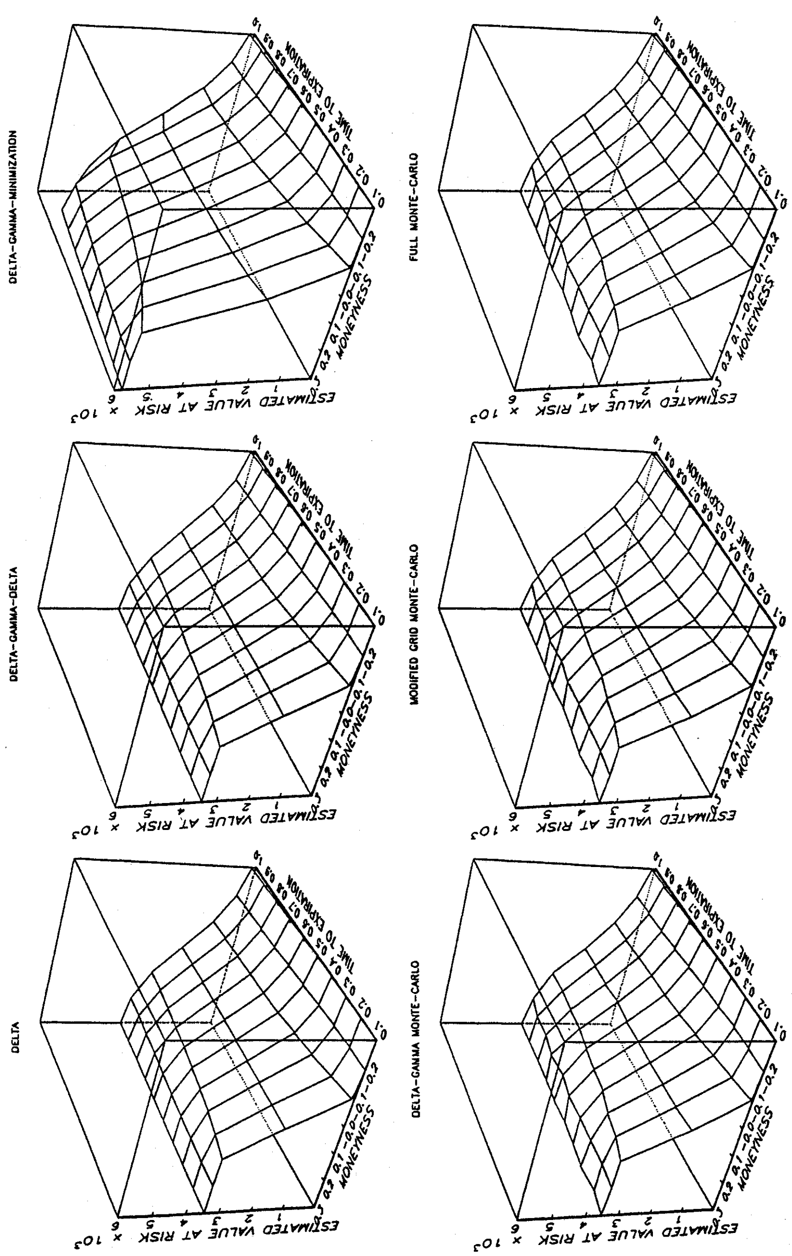


Figure 1. 1% Quantile VAR estimates by VAR method, moneyness, and time to expiration.

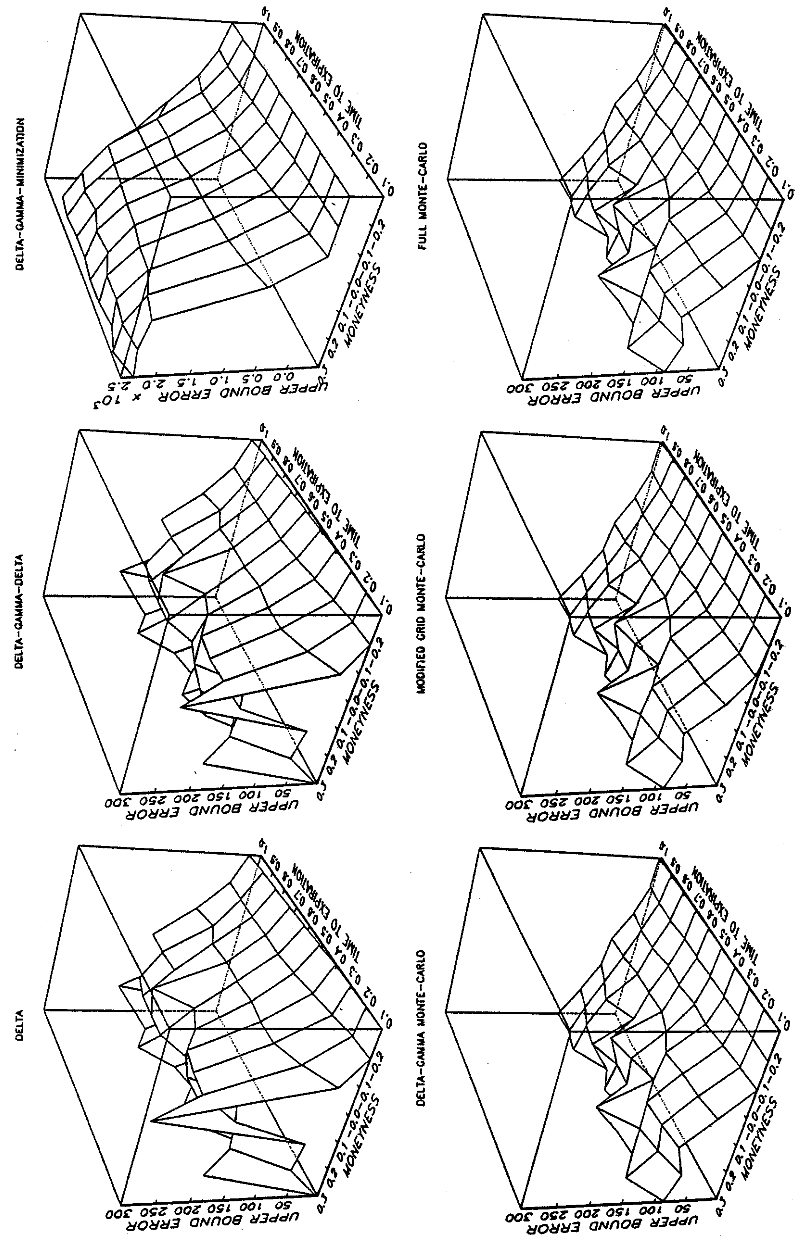
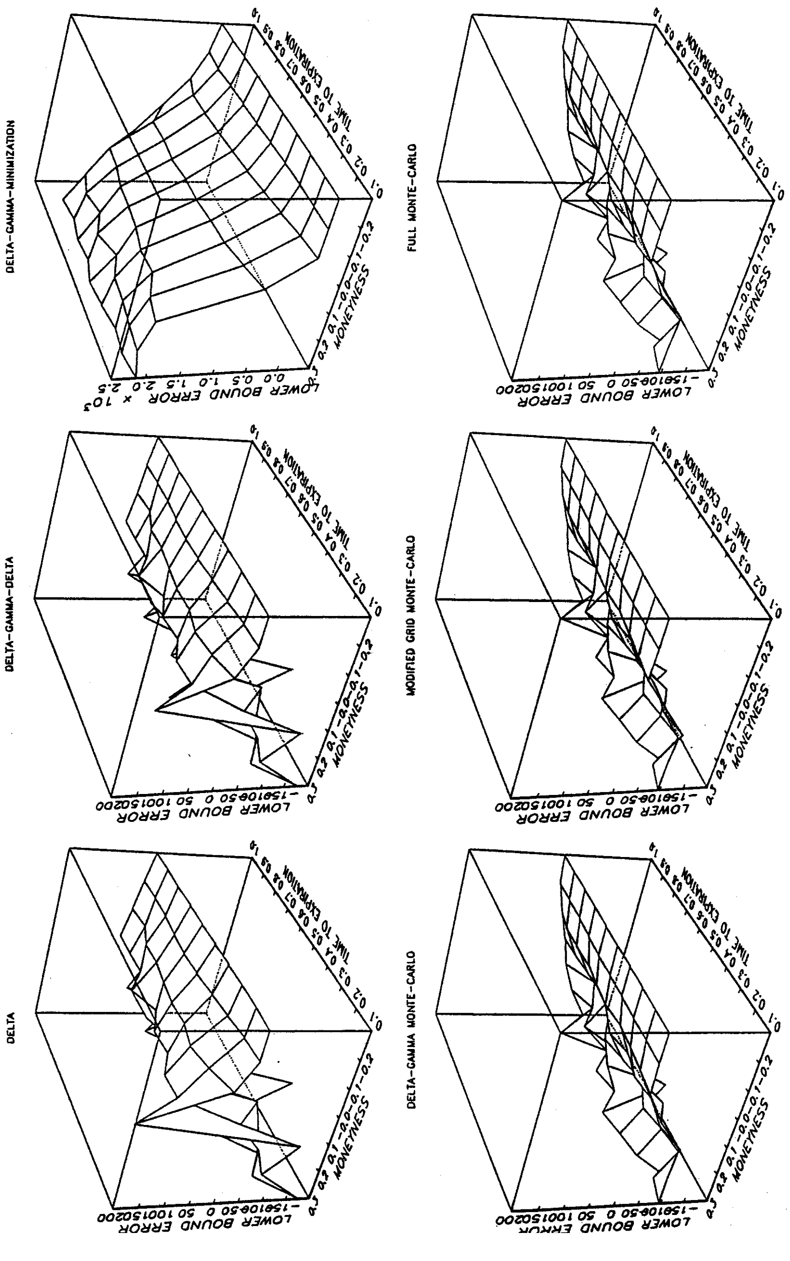


Figure 2. Upper Bound Errors in 1% quantile VAR estimates by VAR method, moneyness, and time to expiration.



Figcap 3. Lower Bound Errors in 1% quantile VAR estimates by VAR method, moneyness, and time to expiration.

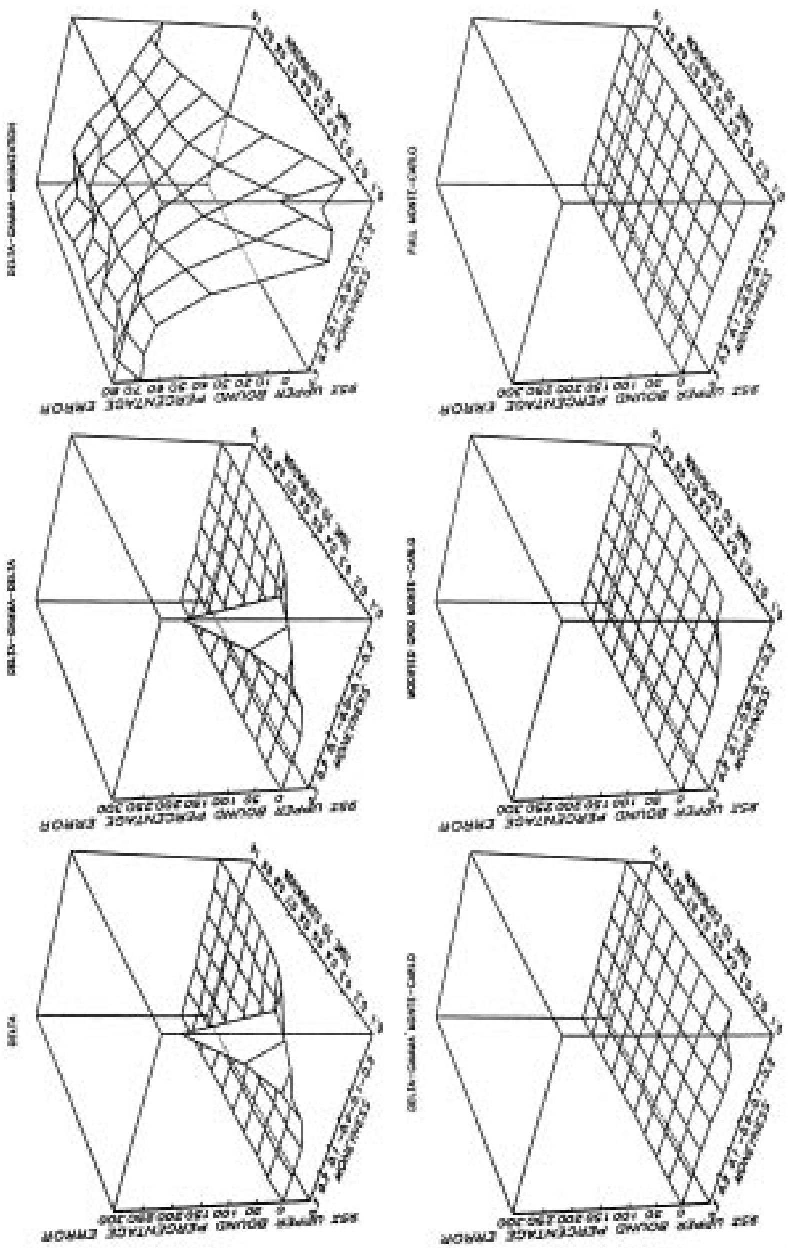


Figure 4. Upper bound 95% conf interval for %ERRS: 1% quantile VAR estimates by VAR method, moneyness and time to expiration.

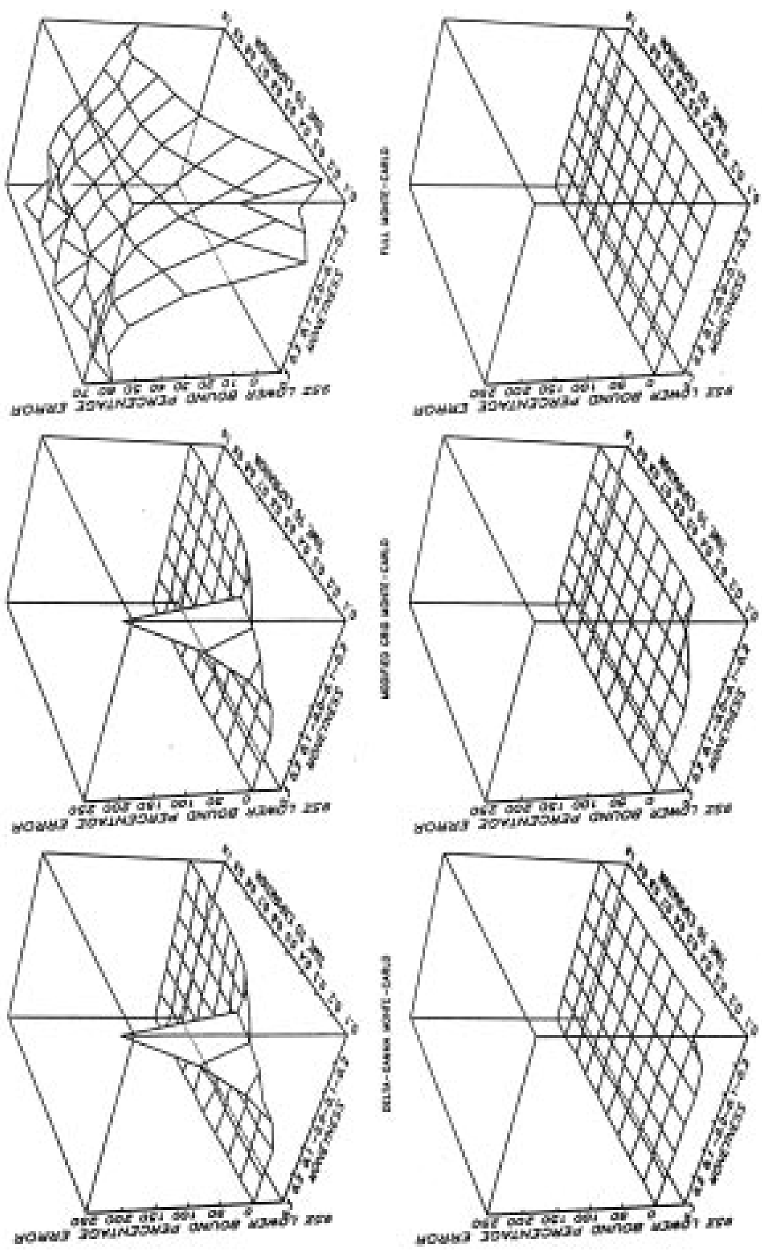


Figure 5. Lower bound 95% conf interval for %ERRS: 1% quantile VAR estimates by VAR method, moneyness and time to expiration.

method appears to be similar; its strange distributional assumptions do not improve much on the results from the delta method. The delta gamma minimization method also overstates VAR, but by very large amounts. By contrast, figures 2–4 show the confidence intervals for the errors from the delta-gamma Monte Carlo method and delta-gamma delta methods usually contain 0. The exception is for long call-options that are deep out-of-the-money and close to expiration. This is not surprising since these options have low values, large exchange rate deltas, and their only risk is that option prices could decline. Since they have large deltas, but the option prices cannot decline much further, the dynamics of the option prices must be highly nonlinear; this nonlinearity is not adequately captured by any of the methods except full Monte Carlo. However, the delta-gamma Monte Carlo and modified grid Monte Carlo do a much better job with these nonlinearities than do the delta and delta-gamma-delta methods.

The results for the *short call-option positions* (not presented) for VAR at the 1% confidence level are qualitatively similar. The delta and delta-gamma-delta methods tend to understate value at risk for near the money options, because the exchange rate delta is increasing in the underlying exchange rate, and the delta method does not take this into account. The delta-gamma-minimization method continues to substantially overstate VAR. Finally, the delta-gamma Monte Carlo method and the modified grid Monte Carlo method overstate VAR slightly for deep out-of-the-money options with a short time to maturity.

Table 3 provides a large amount of supplemental information on the results in figures 1–5. For those portfolios where a VAR method was statistically detected overstating VAR, panel A present the proportion (FREQ) of portfolios for which VAR was overstated, and the mean and standard deviation (STD) of overstatement conditional on overstatement using three measures of overstatement,[42] LOW, MEDIUM, and HIGH. LOW is the most optimistic measure of error and HIGH is the most pessimistic. LOW and HIGH are endpoints of a 95% confidence interval for the amount of overstatement; MEDIUM is always between LOW and HIGH and is the difference between a VAR estimate and a full Monte Carlo estimate.

Panel B provides analogous information to panel A but uses the methods in Appendix 1 to scale the results as a per cent of true VAR. This facilitates comparisons of results across the VAR literature and makes it possible to interpret the results in terms of risk-based capital. For example, suppose risk-based capital charges are proportional to estimated VAR. A firm uses the Delta method to compute VAR and to determine its risk based capital, and its entire portfolio is a long position in one of these call options. In these (extreme) circumstances, for 62.86% of the long call-option positions, VAR at the 1% confidence level would be overstated, and the firm's risk-based capital would exceed required capital by an average of 24.66% (LOW) to 30.50% (HIGH) with a standard deviation of about 45%, meaning that for some portfolios risk-based capital could exceed required capital by 100%. It is important to exercise caution when interpreting these percentage results; the errors in percentage terms should be examined with the errors in levels. For example, for the long call-option positions, the largest percentage errors occur for deep out-of-the-money options. Since VAR (based on full Monte Carlo estimates) and estimates of VAR errors in levels are low for these options, a large percentage error is not important unless these options are a large part of a firm's portfolio.

Table 3. Summary statistics for simulation exercise 1.

Long Call, 1 % Quantile VAR Estimates:
A. Overstatement of VAR

VAR method	Error measure	Mean	STD	FREQ	MAE	RMSE
	LOW	37.13	37.33	62.86	23.34	41.75
Delta	MEDIUM	61.96	55.07	62.86	38.95	65.72
	HIGH	83.57	75.69	62.86	52.53	89.39
	LOW	28.36	29.18	61.43	17.42	31.89
DCD	MEDIUM	51.48	47.02	61.43	31.62	54.64
	HIGH	71.94	67.30	61.43	44.19	77.21
	LOW	1043.26	868.29	95.71	998.55	1327.92
DGMin	MEDIUM	1102.26	913.93	95.71	1055.02	1400.84
	HIGH	1152.28	952.66	95.71	1102.89	1462.70
	LOW	0.00	0.00	2.86	0.01	0.07
DGMC	MEDIUM	0.00	0.00	2.86	0.02	0.14
	HIGH	0.00	0.00	2.86	0.03	0.25
	LOW	0.34	0.72	10.00	0.03	0.25
ModGridMC	MEDIUM	0.51	1.02	10.00	0.05	0.36
	HIGH	0.62	1.20	10.00	0.06	0.43
	LOW	0.00	0.00	0.00	0.00	0.00
FullMC	MEDIUM	0.00	0.00	0.00	0.00	0.00
	HIGH	0.00	0.00	0.00	0.00	0.00

B. Overstatement as % of VAR

VAR method	Error measure	Mean	STD	FREQ	MAE	RMSE
	LOW	24.66	45.82	62.86	15.50	41.25
Delta	MEDIUM	27.69	45.53	62.86	17.40	42.25
	HIGH	30.50	45.20	62.86	19.17	43.23
	LOW	22.41	44.88	61.43	13.76	39.32
DGD	MEDIUM	25.36	44.60	61.43	15.58	40.21
	HIGH	28.12	44.25	61.43	17.28	41.10
	LOW	47.62	15.15	95.71	45.63	48.89
DGMin	MEDIUM	52.09	16.20	95.71	49.92	53.37
	HIGH	56.19	17.17	95.71	53.87	57.48
	LOW	16.29	14.66	2.86	0.55	3.74
DGMC	MEDIUM	16.99	14.49	2.86	0.63	3.85
	HIGH	17.57	14.30	2.86	0.73	3.99
	LOW	12.71	15.14	10.00	1.27	6.25
ModGridMC	MEDIUM	14.23	14.70	10.00	1.42	6.47
	HIGH	15.59	14.25	10.00	1.56	6.68
	LOW	0.00	0.00	0.00	0.00	0.00
FullMC	MEDIUM	0.00	0.00	0.00	0.00	0.00
	HIGH	0.00	0.00	0.00	0.00	0.00

Table 3. (continued)

Long Call, 1 % Quantile VAR Estimates:
C. Understatement of VAR

VAR method	Error measure	Mean	STD	FREQ
	LOW	0.00	0.00	0.00
Delta	MEDIUM	0.00	0.00	0.00
	HIGH	0.00	0.00	0.00
	LOW	0.00	0.00	0.00
DGD	MEDIUM	0.00	0.00	0.00
	HIGH	0.00	0.00	0.00
	LOW	0.00	0.00	1.43
DGMin	MEDIUM	0.00	0.00	1.43
	HIGH	0.00	0.00	1.43
	LOW	0.22	0.26	4.29
DGMC	MEDIUM	0.44	0.50	4.29
	HIGH	0.78	0.89	4.29
	LOW	0.00	0.00	0.00
ModGridMC	MEDIUM	0.00	0.00	0.00
	HIGH	0.00	0.00	0.00
	LOW	0.00	0.00	0.00
FullMC	MEDIUM	0.00	0.00	0.00
	HIGH	0.00	0.00	0.00

D. Understatement as % of VAR

VAR method	Error measure	Mean	STD	FREQ
	LOW	0.00	0.00	0.00
Delta	MEDIUM	0.00	0.00	0.00
	HIGH	0.00	0.00	0.00
	LOW	0.00	0.00	0.00
DGD	MEDIUM	0.00	0.00	0.00
	HIGH	0.00	0.00	0.00
	LOW	3.48	0.00	1.43
DGMin	MEDIUM	4.88	0.00	1.43
	HIGH	6.53	0.00	1.43
	LOW	2.08	1.13	4.29
DGMC	MEDIUM	3.48	1.14	4.29
	HIGH	5.40	0.95	4.29
	LOW	0.00	0.00	0.00
ModGridMC	MEDIUM	0.00	0.00	0.00
	HIGH	0.00	0.00	0.00
	LOW	0.00	0.00	0.00
FullMC	MEDIUM	0.00	0.00	0.00
	HIGH	0.00	0.00	0.00

Table 3. (continued)

Long Call, 1 % Quantile VAR Estimates:
E. VAR errors statistically equal to 0

VAR method	Error measure	Mean	STD	FREQ
	LOW	– 92.69	62.79	37.14
Delta	MEDIUM	17.35	50.95	37.14
	HIGH	109.71	50.12	37.14
	LOW	– 92.46	62.56	38.57
DGD	MEDIUM	17.15	50.82	38.57
	HIGH	108.71	49.47	38.57
	LOW	– 0.77	0.65	2.86
DGMin	MEDIUM	– 0.26	0.27	2.86
	HIGH	0.07	0.04	2.86
	LOW	– 60.79	50.41	92.86
DGMC	MEDIUM	0.02	0.58	92.86
	HIGH	51.58	40.71	92.86
	LOW	– 64.64	50.85	90.00
ModGridMC	MEDIUM	– 1.90	2.40	90.00
	HIGH	51.29	39.41	90.00
	LOW	– 56.48	50.98	100.00
FullMC	MEDIUM	0.00	0.00	100.00
	HIGH	47.88	41.42	100.00

F. VAR errors statistically equal to 0 as % of VAR

VAR method	Error measure	Mean	STD	FREQ
	LOW	– 2.88	1.89	37.14
Delta	MEDIUM	0.64	1.70	37.14
	HIGH	3.62	1.83	37.14
	LOW	– 2.88	1.88	38.57
DGD	MEDIUM	0.63	1.67	38.57
	HIGH	3.59	1.78	38.57
	LOW	– 2.50	0.72	2.86
DGMin	MEDIUM	– 0.48	0.71	2.86
	HIGH	0.90	0.85	2.86
	LOW	– 3.22	0.82	92.86
DGMC	MEDIUM	– 0.09	0.24	92.86
	HIGH	2.63	0.77	92.86
	LOW	– 3.14	1.04	90.00
ModGridMC	MEDIUM	0.04	0.43	90.00
	HIGH	2.79	0.58	90.00
	LOW	– 3.01	0.98	100.00
FullMC	MEDIUM	0.00	0.00	100.00
	HIGH	2.60	0.75	100.00

Notes. For VAR at the 1% confidence level, for long call-option positions (from simulation 1), the table provides summary statistics on six methods for computing VAR using three measures of VAR error or percentage error. The option positions are long one call option in one of 70 U.S. dollar/French franc options that deliver 1 million French francs if exercised. The options vary by moneyness and time to expiration. The VAR methods are Delta, Delta-Gamma-Delta (DGD), Delta-Gamma-Minimization (DGMin), Delta-Gamma Monte Carlo (DGMC), Modified Grid Monte Carlo (ModGridMC), and Monte Carlo with full repricing (FullMC). For each group of option positions, the table reports the proportion (FREQ) for which VAR was overstated, understated, or statistically indistinguishable from zero. For positions classified as over- or understated, the Mean and Standard Deviation (STD) of over- or understatement are computed for three measures of error, LOW (an optimistic measure of error), MEDIUM (a less optimistic measure of error), and HIGH (a pessimistic measure of error). LOW and HIGH are the endpoints of a 95% confidence interval for the amount of over- or under statement. For positions classified as statistically indistinguishable from zero, LOW and HIGH are endpoints of a 95% confidence interval for VAR error. For all positions MEDIUM is always between LOW and HIGH, and measures error as the difference between a VAR estimate and FullMC estimate or as the difference between a VAR estimate and FullMC as a percentage of FullMC. A VAR estimate was classified as overstated if the 95% confidence interval for its error was bounded below by 0; it was classified as understated if the confidence interval for its error was bounded above by 0, and was statistically indistinguishable from 0 if the 95% confidence interval for its error contained 0. To rank the VAR methods two statistical loss functions are reported. Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). These loss functions were computed using an optimistic measure of error (LOW), a pessimistic measure (HIGH), and a middle range measure (MEDIUM). The loss function results in panel A are computed based on the levels of errors, the loss function results in panel B are computed based on the errors as a per cent of true VAR. Details on the VAR methods are contained in section 2; details on the confidence intervals are contained in appendix1; details on the option positions and statistical loss functions are contained in section 5 and Appendix 2.

Panels C and D are analogous to A and B but present results for understatement of VAR. Panels E and F present results on those VAR estimates for which its error was statistically indistinguishable from 0. In this case, LOW and HIGH are endpoints of a 95% confidence bound for the error, which can be negative (LOW) or positive (HIGH). MEDIUM is computed as before. The purpose of panels E and F is to give an indication of the size of error for those VAR errors which were statistically indistinguishable from 0. The panels show percentage errors for VAR estimates which were statistically indistinguishable from 0 probably ranged within plus or minus 3% of VAR.

The conditional distributions present useful information on the VAR method's implications for capital adequacy, but can be misleading when trying to rank the methods.[43] To compare the methods, panels A and B report two statistical loss functions, Mean Absolute Error (MAE), the average absolute error for each VAR method, and Root Mean Squared Error (RMSE), the average squared errors for each VAR method.[44] The errors in these loss functions are computed using the Monte Carlo draws from the full Monte Carlo method. Hence, they are not meaningful for the full Monte Carlo method, but are meaningful for other methods.

Table 3 confirms basic features of the figures. The delta-gamma-minimization method overstates VAR by very large amounts; and the delta and delta-gamma-delta methods tend to overstate VAR for the long call-option positions, and tend to understate it for short option positions (not shown). The table also shows the frequency of under- and overstatement is much lower for the delta-gamma Monte Carlo and modified grid Monte Carlo methods than for other methods. Finally, by all three error measures, the statistical loss functions in table 3 show the best method is delta-gamma Monte Carlo, followed by modified grid Monte Carlo, delta-gamma-delta, delta, and delta-gamma minimization.

The results in table 4 are analogous to those in table 3, but are provided for simulation exercise 2.[45] Based on frequency of under- or overstatement, the delta-gamma Monte Carlo and modified grid Monte Carlo methods under-or overstate VAR for about 25% of portfolios, at the 1% confidence level. This is about half the frequency of the delta or delta-gamma-delta methods. Despite the large differences in frequency of under- or overstatement, the statistical loss functions across the four methods are much more similar than in table 3. I suspect the increased similarity between the methods is because on a portfolio-wide basis, the errors that delta or delta-gamma-delta methods make for long gamma positions in some options is partially offset by the errors made for short gamma positions in other options. Closer inspection of the results (not shown) revealed more details about the similarity of the loss functions. Roughly speaking, for the portfolios where the delta-gamma Monte Carlo and grid Monte Carlo methods made large errors, the other methods made large errors too. However, for the portfolios where the delta and delta-gamma-delta methods made relatively small errors, the delta-gamma Monte Carlo and modified grid Monte Carlo methods often made no errors. Thus, in table 4, the superior methods are superior mostly because they make fewer small errors. This shows up as a large difference in frequency of errors, but relatively small differences in the statistical loss functions. Because of the smaller differences in the statistical loss functions, the rankings of the methods are a bit more blurred, but still roughly similar: the delta-gamma Monte Carlo and modified grid Monte Carlo methods have similar accuracies that are a bit above the accuracies of the other two methods. Importantly, the best methods in simulation exercise 2 over- or understated VAR for about 25% of the portfolios considered by an average amount of about 10% of VAR with a standard deviation of the same amount.

To briefly summarize, the rankings of the methods in terms of accuracies are somewhat robust across the two sets of simulations. The most accurate methods are delta-gamma Monte Carlo and modified grid Monte Carlo followed by delta-gamma-delta and then delta, and then (far behind) by delta-gamma minimization. Although our results on errors are measured in terms of per cent of true VAR, and thus are relatively easy to interpret, they still have the shortcoming that they are specific to the portfolios we generated. A firm, in evaluating the methods here, should repeat the comparisons of these VAR methods using portfolios that are more representative of its actual trading book.

5.3. Empirical results: computational time

To examine computation time, I conducted a simulation exercise in which I computed VAR for a randomly chosen portfolio of 50 foreign exchange options using all six VAR methods. The results are summarized in table 5. The time for the various methods was increasing in the number of complex computations, as expected, with the exception of the modified grid Monte Carlo method. The different results for the modified grid Monte Carlo method are not too surprising since the complexity of the programming to implement this method is likely to have slowed it down.[46] The delta method took 0.07 seconds, the delta-gamma-delta method required 1.17 seconds, the delta-gamma-minimization method

Table 4. Summary statistics for simulation exercise 2

A. Overstatement of VAR

VAR method	Error measure	Mean	STD	FREQ	MAE	RMSE
	LOW	9285.98	13829.98	22.60	5158.24	12983.17
Delta	MEDIUM	14143.25	15827.45	22.60	7820.36	16638.92
	HIGH	18271.80	18308.28	22.60	10480.18	20712.29
	LOW	9348.09	14306.30	21.20	4773.82	12692.74
DGD	MEDIUM	14491.12	16174.45	21.20	7341.28	16265.99
	HIGH	18856.94	18615.27	21.20	9893.22	20284.14
	LOW	414981.5	356880.8	100.00	414981.5	547333.1
DGMin	MEDIUM	421121.9	361356.2	100.00	421121.9	554907.2
	HIGH	426591.5	365546.4	100.00	426591.5	561786.9
	LOW	15236.24	19463.43	11.60	3954.29	12482.32
DGMC	MEDIUM	20912.81	21023.91	11.60	5545.61	15394.13
	HIGH	25865.86	23197.04	11.60	7200.20	18764.70
	LOW	14640.18	19397.77	12.00	3932.56	12460.24
ModGridMC	MEDIUM	20357.90	20898.54	12.00	5541.86	15374.71
	HIGH	25500.28	23054.59	12.00	7229.88	18785.42
	LOW	0.00	0.00	0.00	0.00	0.00
FullMC	MEDIUM	0.00	0.00	0.00	0.00	0.00
	HIGH	0.00	0.00	0.00	0.00	0.00

B. Overstatement as % of VAR

VAR method	Error measure	Mean	STD	FREQ	MAE	RMSE
	LOW	7.55	8.46	22.60	3.57	8.01
Delta	MEDIUM	11.10	8.95	22.60	5.34	9.96
	HIGH	14.31	9.18	22.60	7.24	12.22
	LOW	7.27	8.85	21.20	3.07	7.05
DGD	MEDIUM	10.84	9.36	21.20	4.72	8.93
	HIGH	14.05	9.61	21.20	6.49	11.10
	LOW	204.36	59.53	100.00	204.36	212.86
DGMin	MEDIUM	214.85	62.05	100.00	214.85	223.63
	HIGH	224.50	63.97	100.00	224.50	233.44
	LOW	10.36	11.64	11.60	2.18	6.63
DGMC	MEDIUM	14.22	12.25	11.60	3.08	8.07
	HIGH	17.64	12.52	11.60	4.01	9.64
	LOW	9.97	11.82	12.00	2.18	6.65
ModGridMC	MEDIUM	13.82	12.42	12.00	3.07	8.09
	HIGH	17.27	12.68	12.00	4.01	9.66
	LOW	0.00	0.00	0.00	0.00	0.00
FullMC	MEDIUM	0.00	0.00	0.00	0.00	0.00
	HIGH	0.00	0.00	0.00	0.00	0.00

Table 4. (continued)

C. Understatement of VAR

VAR method	Error measure	Mean	STD	FREQ
	LOW	10478.12	15896.63	29.20
Delta	MEDIUM	15835.58	18672.39	29.20
	HIGH	21749.17	21870.54	29.20
	LOW	10340.84	16138.02	27.00
DGD	MEDIUM	15811.71	18963.82	27.00
	HIGH	21835.36	22266.65	27.00
	LOW	0.00	0.00	0.00
DGMin	MEDIUM	0.00	0.00	0.00
	HIGH	0.00	0.00	0.00
	LOW	16567.30	19208.84	13.20
DGMC	MEDIUM	23634.26	21539.65	13.20
	HIGH	31816.39	24380.80	13.20
	LOW	16736.45	19209.47	13.00
ModGridMC	MEDIUM	23837.82	21549.07	13.00
	HIGH	32075.74	24389.10	13.00
	LOW	0.00	0.00	0.00
FullMC	MEDIUM	0.00	0.00	0.00
	HIGH	0.00	0.00	0.00

D. Understatement as % of VAR

VAR method	Error measure	Mean	STD	FREQ
	LOW	6.37	8.92	29.20
Delta	MEDIUM	9.69	9.43	29.20
	HIGH	13.74	9.95	29.20
	LOW	5.67	6.98	27.00
DGD	MEDIUM	8.97	7.34	27.00
	HIGH	12.99	7.73	27.00
	LOW	0.00	0.00	0.00
DGMin	MEDIUM	0.00	0.00	0.00
	HIGH	0.00	0.00	0.00
	LOW	7.49	7.96	13.20
DGMC	MEDIUM	10.81	8.21	13.20
	HIGH	14.86	8.51	13.20
	LOW	7.54	7.89	13.00
ModGridMC	MEDIUM	10.86	8.14	13.00
	HIGH	14.93	8.43	13.00
	LOW	0.00	0.00	0.00
FullMC	MEDIUM	0.00	0.00	0.00
	HIGH	0.00	0.00	0.00

Table 4. (continued)

E. VAR errors statistically equal to 0

VAR method	Error measure	Mean	STD	FREQ
	LOW	− 7035.28	7256.65	48.20
Delta	MEDIUM	− 155.78	4747.76	48.20
	HIGH	6010.55	5747.92	48.20
	LOW	− 6896.76	6904.83	51.80
DGD	MEDIUM	− 287.25	4330.18	51.80
	HIGH	5633.43	5374.26	51.80
	LOW	0.00	0.00	0.00
DGMin	MEDIUM	0.00	0.00	0.00
	HIGH	0.00	0.00	0.00
	LOW	− 5872.74	5580.88	75.20
DGMC	MEDIUM	− 19.14	2639.36	75.20
	HIGH	5249.76	4947.50	75.20
	LOW	− 5922.22	5606.95	75.00
ModGridMC	MEDIUM	− 77.73	2636.13	75.00
	HIGH	5161.39	4810.13	75.00
	LOW	− 6140.43	4897.27	100.00
FullMC	MEDIUM	0.00	0.00	100.00
	HIGH	5469.60	4576.15	100.00

F. VAR errors statistically equal to 0 as % of VAR

VAR method	Error measure	Mean	STD	FREQ
	LOW	− 3.53	1.98	48.20
Delta	MEDIUM	− 0.03	1.78	48.20
	HIGH	3.06	1.74	48.20
	LOW	− 3.58	1.88	51.80
DGD	MEDIUM	− 0.08	1.65	51.80
	HIGH	2.99	1.60	51.80
	LOW	0.00	0.00	0.00
DGMin	MEDIUM	0.00	0.00	0.00
	HIGH	0.00	0.00	0.00
	LOW	− 3.48	1.24	75.20
DGMC	MEDIUM	0.02	0.89	75.20
	HIGH	3.08	1.12	75.20
	LOW	− 3.51	1.22	75.00
ModGridMC	MEDIUM	− 0.01	0.93	75.00
	HIGH	3.05	1.19	75.00
	LOW	− 3.54	0.85	100.00
FullMC	MEDIUM	0.00	0.00	100.00
	HIGH	3.07	0.69	100.00

Notes. For VAR at the 1% confidence level, for 500 randomly chosen portfolios of options (from simulation 2), the table provides summary statistics on six methods for computing VAR using three measures of VAR error or percentage error. The VAR methods are Delta, Delta-Gamma-Delta (DGD), Delta-Gamma-Minimization (DGMin), Delta-Gamma Monte Carlo (DGMC), Modified Grid Monte Carlo (ModGridMC), and Monte Carlo with full repricing (FullMC). For each group of option positions, the table reports the proportion (FREQ) for with VAR was overstated, understated, or statistically indistinguishable from zero. For positions classified as over- or understated, the Mean and Standard Deviation (STD) of over- or understatement are computed for three measures of error, LOW (an optimistic measure of error), MEDIUM (a less optimistic measure of error), and HIGH (a pessimistic measure of error). LOW and HIGH are the endpoints of a 95% confidence interval for the amount of over- or understatement. For positions classified as statistically indistinguishable from zero, LOW and HIGH are endpoints of a 95% confidence interval for VAR error. For all positions MEDIUM is always between LOW and HIGH, and measures error as the difference between a VAR estimate and FullMC estimate or as the difference between a VAR estimate and FullMC as a percentage of FullMC. A VAR estimate was classified as overstated if the 95% confidence interval for its error was bounded below by 0; it was classified as understated if the confidence interval for its error was bounded above by 0, and was statistically indistinguishable from 0 if the 95% confidence interval for its error contained 0. To rank the VAR methods two statistical loss functions are reported, Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE). These loss functions were computed using an optimistic measure of error (LOW), a pessimistic measure (HIGH), and a middle range measure (MEDIUM). The loss function results in panel A are computed based on the levels of errors, the loss function results in panel B are computed based on the errors as a per cent of true VAR. Details on the VAR methods are contained in section 2, details on the confidence intervals are contained in appendix 1; details on the option positions and statistical loss functions are contained in section 5 and Appendix 2.

required 1.27 seconds, the delta-gamma Monte Carlo method required 3.87 seconds, the grid Monte Carlo method required 32.29 seconds, and full Monte Carlo required 66.27 seconds. This translates into 0.003 hours per VAR computation for a portfolio of 10,000 options using the delta method, 0.065 hours using the delta-gamma-delta method, 0.07 hours using the delta-gamma minimization method, 0.215 hours using the delta-gamma Monte Carlo method, 1.79 hours using the modified grid Monte Carlo method, and 3.68 hours using full Monte Carlo. Of course, the relative figures are more important than the literals since these figures depend on the computer technology at the Federal Reserve Board. The results on time that are reported here were computed using Gauss for Unix on a Spacc 20 workstation.

5.4. Accuracy versus computational time

This article investigated six methods of computing Value at Risk for both accuracy and computational time. The results for full Monte Carlo were the most accurate, but also required the largest amount of time to compute. The next most accurate methods were the delta-gamma Monte Carlo and modified grid Monte Carlo methods. Both attained comparable levels of accuracy, but the delta gamma Monte Carlo method is faster by more than a factor of 8. The next most accurate methods are the delta-gamma-delta and delta methods. These methods are faster than the delta-gamma Monte Carlo methods by a factor of 3 and 51 respectively. Finally, the delta-gamma minimization method is slower than the delta-gamma delta method and is the most inaccurate of the methods.

Based on these results, the delta-gamma minimization method seems to be dominated by other methods which are both faster and more accurate. Similarly, the modified grid

Table 5. Results on computational time.

VAR method	Computation time (seconds)	# Complex computations	# Simple computations
Delta	0.076	504	1
DGD	1.17	3300	1
DGMIN	1.27	3300	1
DGMC	3.88	3300	10000
ModGridMC	32.29	304	10000
FullMC	66.27	10000	0

Notes. The table presents information on the time required, and the computational complexity involved, for computing VAR for a portfolio that contained 50 European foreign exchanged options using six different methods of computing Value at Risk. Forty-eight of the options were mapped to five risk factors in RiskMetrics and two options were mapped to six risk factors. The VAR methods are the Delta, Delta-Gamma-Delta (DGD), Delta-Gamma-Minimization (DGMIN). Delta-Gamma Monte Carlo, Modified Grid Monte Carlo andf Full Monte Carlo methods. Details on the methods are contained in section 2. Details on computational requirements are contained in section 4.

Monte Carlo method is dominated by the delta-gamma Monte Carlo method, since the latter method has the same level of accuracy but can be estimated more rapidly. The remaining four methods require more computational time to attain more accuracy. Thus, the remaining four methods cannot be strictly ranked. However, in my opinion, full Monte Carlo is too slow to produce VAR estimates in a timely fashion, while the delta-gamma Monte Carlo methods is reasonably fast (0.215 hours per 10,000 options), and produces amongst the next most accurate results both for individual options, as in simulation 1, and for portfolios of options as in simulation 2. Therefore, I believe the best of the methods considered here is delta-gamma Monte Carlo.

6. Conclusions

Methods of computing Value at Risk need to be both accurate and available on a timely basis. There is likely to be an inherent tradeoff between these objectives since more rapid methods tend to be less accurate. This article investigated the tradeoff between accuracy and computational time, and it also introduced a method for measuring the accuracy of VAR estimates based on confidence intervals from Monte Carlo with full repricing. The confidence intervals allowed us to quantify VAR errors both in monetary terms and as a per cent of true VAR even though true VAR is not known. This lends a capital adequacy interpretation to the VAR errors for firms that choose their risk based capital based on VAR.

To investigate the tradeoff between accuracy and computational time, this article investigated six methods of computing VAR. The accuracy of the methods varied fairly widely When examining accuracy and computational time together, two methods dominated because other methods were at least as accurate and required a smaller amount of time to compute. Of the undominated methods, the delta-gamma Monte Carlo method was the most accurate of those that required a reasonable amount of computation time. Other advantages of the delta-gamma Monte Carlo method is that it allows the assumption of normally distributed factor shocks to be relaxed in favor of alternative distributional assumptions including historical simulation. Also, the delta-gamma Monte Carlo method can be implemented in a parallel fashion across the firm, i.e., delta and gamma matrices can be computed for each trading desk, and then the results can be aggregated up for a firm-wide VAR computation.

While the delta-gamma Monte Carlo method produces among the most accurate VAR estimates, it still frequently produced errors that were statistically significant and economically large. For 25% of the 500 randomly chosen portfolios of options in simulation exercise 2, it over- or understated VAR by an average of 10% of VAR with a standard deviation of the same amount; and for deep out-of-the-money options in simulation 1, its accuracy was poor. More importantly, its errors when actually used are likely to be different than those reported here for two reasons. First, the results reported here abstract away from errors in specifying the distribution of the factor shocks; these errors are likely to be important in practice. Second, firms' portfolios contain additional types of instruments not considered here, and many firms have hedged positions, while the ones used here are unhedged.

In order to better characterize the methods in terms of accuracy and computational time, it is important to do the type of work performed here using more realistic factor shocks, and a firm's actual portfolio. This article has begun the groundwork to perform this type of analysis.

Appendix 1: Nonparametric confidence intervals

The confidence intervals derived in this section draw on a basic result from the theory of order statistics. To introduce this result, let $X_{(1)} < X_{(2)}, \ldots, < X_{(N)}$ be order statistics from N i.i.d. random draws of the continuous random variable X with unknown distribution function G, and let ξ_p represent the pth percentile of G. The result in the following proposition illustrates how a confidence interval for the pth percentile of G can be formed from order statistics. The proof of this proposition is from the second edition of the book *Order Statistics* by Herbert A. David.

Proposition 1: For $r < s$,

$$\mathrm{Prob}(X_{(r)} < \xi_p < X_{(s)}) = \sum_{i=r}^{s-1} \binom{N}{i} p^i (1-p)^{N-i}$$

Proof:

$$\begin{aligned} r < s \Rightarrow\ & \mathrm{Prob}(X_{(r)} < \xi p) = \mathrm{Prob}(X_{(s)} \leq \xi_p) + \mathrm{Prob}(X_{(r)} < \xi_p < X_{(s)}) \\ \Rightarrow\ & \mathrm{Prob}(X_{(r)} < \xi_p < X_{(s)}) = \mathrm{Prob}(X_{(r)} \leq \xi_p) - \mathrm{Prob}(X_{(s)} \leq \xi_p) \\ =\ & \sum_{i=0}^{N-r} \binom{N}{r+i} p^{r+i} (1-p)^{N-r-i} - \sum_{j=0}^{N-s} \binom{N}{s+j} p^{s+j} (1-p)^{N-s-j} \\ =\ & \sum_{i=r}^{s-1} \binom{N}{i} p^i (1-p)^{N-i} \end{aligned}$$

Confidence intervals from Monte Carlo estimates

To construct a 95% confidence interval for the pth percentile of G using the results from Monte Carlo simulation, it suffices to solve for an r and s such that

$$\sum_{i=r}^{s-1}\binom{N}{i}p^i(1-p)^{N-i} \geq 0.95.$$

and such that

$$\sum_{i=r+1}^{s-1}\binom{N}{i}p^i(1-p)^{N-i} \leq 0.95.$$

Then the order statistics $X_{(r)}$ and $X_{(s)}$ from the Monte Carlo simulation are the bounds for the confidence interval. In general, there are several r and s pairs that satisfy the above criteria. The confidence intervals contained in table 1 impose the additional restriction that the confidence interval be as close to symmetric as possible about the pth percentile of the Monte Carlo distribution, i.e., r and s were chosen so that $p - r/N \approx s/N - p$. No more than two r and s pairs can satisfy all three criteria. In circumstances where there were two r and s pairs to choose from, I chose one randomly.

Finally, since value at risk at confidence level p corresponds to the pth percentile of some unknown distribution, the above results make it possible to create confidence intervals for Value at Risk based on the Monte Carlo results.

Confidence intervals for VAR errors

The bounds from the above Monte Carlo confidence intervals can be used to form confidence intervals for the errors from a VAR estimate. More specifically, let X_T denote true Value at Risk, and let X_H and X_L denote bounds such that

$$\text{Prob}(X_L \geq X_T \geq X_H) = 0.95.$$

From the analysis above we know we can find these bounds. Given these bounds, let $\hat{X}$ be any other estimate of Value at Risk. It then follows that

$$\text{Prob}(\hat{X} - X_L \leq \hat{X} - X_T \leq \hat{X} - X_H) = 0.95.$$

Therefore, $\hat{X} - X_L$ and $\hat{X} - X_H$ form upper and lower bounds for a confidence interval for the error when using VAR estimate $\hat{X}$.

The magnitude of the VAR errors made in any particular portfolio depends on the size of the positions in the portfolio. Therefore, in ''small'' portfolios, all methods may generate small VAR errors, and thus appear similar when in fact the methods are very different. The problem is that the VAR errors need to be appropriately scaled when comparing the methods. Perhaps the best way to scale the errors, for purposes of comparison, is to scale the errors as a percent of true VAR, i.e., to examine VAR percentage errors. The construction of confidence intervals for VAR Percentage Errors is discussed in the next section.

Confidence Intervals for VAR percentage errors

The bounds from the above Monte Carlo confidence intervals can be used to form confidence intervals for the percentage errors made when using various methods of computing Value at Risk. To create this confidence interval requires some notation:

Let X_T denote the true Value at Risk, and let A denote the indicator function

$$A = 1_{\{L \le X_T \le H\}}$$

where L and H are bounds such that $H > L > 0$.[47]

Similarly, let $\hat{X}$ be an estimator of Value at Risk, with percentage error denoted by $\%err\hat{X}$ where $\%err\hat{X} = 100\frac{\hat{X}-X_T}{X_T}$, and let $A*$ be the indicator function

$$A^* = 1_{\{L^* \le \%err\hat{X} \le H^*\}},$$

where

$$H^* = \sup\left(100\frac{\hat{X}-L}{L}, 100\frac{\hat{X}-L}{H}\right),$$

and

$$L^* = \inf\left(100\frac{\hat{X}-H}{H}, 100\frac{\hat{X}-H}{L}\right).$$

$L*$ and $H*$ are bounds for the percentage error in calculating Value at Risk using $\hat{X}$. The probability that $\hat{X}$ is in these bounds is given by the following proposition:

Proposition 2: If L and H are greater than 0, and bound true Value at Risk with probability q, then $L*$ and $H*$ bound %err$\hat{X}$ with probability greater than q.

Proof. In mathematical terms, the proposition says

$$\mathrm{Prob}(\mathrm{A} = 1) = \mathrm{q}, \Rightarrow \mathrm{Prob}(\mathrm{A}^* = 1) \ge \mathrm{q}.$$

The main part of the proof involves showing that $\{A = 1\} \Rightarrow \{A^* = 1\}$. If this is true it follows that

$$\mathrm{Prob}(\mathrm{A}^* = 1|\mathrm{A} = 1) = 1.$$

Furthermore,

$$\begin{aligned}\mathrm{Prob}(A^* = 1) &= \mathrm{Prob}(A^* = 1|A = 1)\mathrm{Prob}(A = 1) + \mathrm{Prob}(A* = 1|A = 0)\mathrm{Prob}(A = 0)\\ &\ge \mathrm{Prob}(A^* = 1|A = 1)\mathrm{Prob}(A = 1) = q.\end{aligned}$$

To prove $\{A = 1\} \Rightarrow \{A^* = 1\}$, suppose that $A = 1$, *i.e.* :

$$0 < L < X_T < H.$$

By algebra it follows that:

$$\frac{\hat{X}-L}{X_T} > \frac{\hat{X}-X_T}{X_T} > \frac{\hat{X}-H}{X_T}. \tag{A1}$$

There are three cases to consider:

Case 1: $\hat{X} > H$ and $A = 1$. In this case, all terms in (A1) are positive. If the first term in the inequality is multiplied by a positive number greater than 1, and the last term is multiplied by a positive number less than 1, the inequalities will still hold. Since $X_T/L > 1$ and $X_T/H < 1$, multiplying terms 1 and 3 by these respectively shows: $\hat{X}>H \textit{ and } A = 1 \Rightarrow$

$$\frac{\hat{X}-L}{L} > \frac{\hat{X}-X_T}{X_T} > \frac{\hat{X}-H}{H}. \tag{A1}$$

Case 2: $\hat{X} < L$ and $A = 1$. In this case all terms in (A1) are negative. If the first term in (1) is multiplied by a positive number less than 1, and the last term is multiplied by a positive number greater than 1, the inequalities will still hold. Since $X_T/H < 1$ and $X_T/L > 1$, multiplying terms 1 and 3 by these respectively shows: $\hat{X} < H$ and $A = 1 \Rightarrow$

$$\frac{\hat{X}-L}{H} > \frac{\hat{X}-X_T}{X_T} > \frac{\hat{X}-H}{L}. \tag{A1}$$

Case 3: $L < \hat{X} < H$ and $A = 1$. In this case, the first term in (A1) is positive and the last is negative. Therefore if the first term in (1) is multiplied by a positive number greater than 1, and the last term is multiplied by a positive number greater than 1, then the inequalities will still hold. Since $X_T/L > 1$, multiplying terms 1 and 3 by this shows: $\hat{X} < H$ and $A = 1 \Rightarrow$

$$\frac{\hat{X}-L}{L} > \frac{\hat{X}-X_T}{X_T} > \frac{\hat{X}-H}{L}.$$

Inspection shows that in Cases 1–3 the upper bound corresponds to the formula for H^* and the lower bound corresponds to the formula for L^*. Therefore, $\{A = 1\} \Rightarrow \{A^* = 1\}$ and thus the proposition is proved.

It immediately follows that if L and H are bounds of a 95% confidence interval for a Monte Carlo estimate of value at risk, then L^* and H^* bound the percentage error of the Value at Risk estimate with confidence exceeding 95%.

Appendix 2: Details on simulations 1 and 2{\rm (a1)

Detail on data

Factor volatilities, correlations, and exchange rates used in the simulations were extracted from JP Morgan's RiskMetrics database for August 17, 1992. Eurointerest rates for the same date were provided by the Federal Reserve Board. The implied volatilities that were used to price each option were the one-day standard deviations of (df/f) from RiskMetrics^{TM}, where $100(df/f)$ represent percentage changes in the relevant exchange rates.[48]

Details on simulation 2

Simulation 2 generated 500 random portfolios of foreign exchange options. The number of options in each portfolio ranged from 1–50 with equal probability. For each option, the amount of foreign currency that was delivered if exercised was chosen from a uniform distribution with support from 1–11 million foreign currency units. Whether an option was a put or a call and whether the portfolio was long or short the option was chosen randomly and with equal probability.

To choose the exchange rate pairs for the options, define each option's home currency as the currency in which its price is denominated and its foreign currency as the currency it delivers if exercised. Similarly, define the home currency in a financial center as the national currency in the center's physical location and define a foreign currency as a currency other than the home currency. For many home currencies, table 9G of the *Central Bank Survey of Foreign Exchange and Derivatives Market Activity* provides information on the amount of turnover in the derivatives market broken down by foreign currency. The amounts of relative turnover in the home and foreign currencies of all currencies considered in the simulation were used as the probability function to generate the home and foreign currency pairs in the simulation with the exception of the U.S. dollar. Because the VAR measurements here are measured in dollars, in all currency pairs where the dollar was chosen as the foreign currency, I switched them and made the dollar the home currency.

Notes for figures

Figures 1 through 5 present results on the properties of six different methods of computing Value at Risk for 70 different option positions. Each position consists of a long position in one call option on the U.S. dollar/French franc exchange rate and delivers 1 million French francs if exercised. The 70 option positions vary by moneyness and time to expiration. Seven moneynesses are examined; they range from 30% out of the money to 30% in the money by increments of 10%. Ten maturities are examined; they range from 0.1 years to 1 year in increments of 0.1 years. The VAR methods that are considered are the delta, delta-gamma-delta, delta-gamma-minimization, delta-gamma Monte Carlo, modified grid Monte Carlo, and full Monte Carlo methods. Details on the methods are contained in section 2 the paper; additional details on the simulation are contained in section 5 and in Appendix 2.

Figure 1 presents estimates of Value at Risk at the 1% quantile; i.e., the figure plots estimates of the 1% quantile of the distribution of the change in option value over a 1 day holding period. Let U and L be upper and lower bounds of a 95% confidence interval for the error made when using VAR estimate $\widehat{\text{VAR}}$ to compute VAR, and let U% and L% be upper and lower bounds of a 95% confidence interval for the *percentage* error made when using VAR estimate $\widehat{\text{VAR}}$ to compute VAR.[49] Figures 2, 3, 4 and 5 present graphs of U, L, U%, and L% respectively for 1% quantile VAR estimates. For those options where U or U% is below zero, $\widehat{\text{VAR}}$ understates VAR with probability exceeding 95%. A very conservative estimate of the amount by which VAR is understated is U, and a very conservative estimate of the amount by which VAR is understated as a per cent of true VAR is U%.[50] Similarly, for those options where L or L% is above zero, $\widehat{\text{VAR}}$ overstates VAR with 95% confidence. Very conservative estimates of the amount of understatement and percentage understatement are L and L%.

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Notes

1. A bibliography of VAR literature that is periodically updated is maintained on the Internet by Barry Schacter at http://pw2.netcom.com/ bschacht/varbiblio.html.
2. If a firm is expected to lose no more than \$10 million over the next day except in 1% of circumstances, then its Value at Risk for a one-day horizon at a 1% confidence level is \$10 million.
3. Boudoukh, Richardson, and Whitelaw (1995) propose a different measure of capital adequacy. They suggest measuring risk as the expected largest loss experienced over a fixed period of time, such as, for example, the expected largest daily loss over a period of 20 days.
4. Hendricks (1996) examined 12 different estimates of VAR and found that the standard deviation of their dispersion around the mean of the estimates is between 10 and 15%, indicating that it would not be uncommon for the VAR estimates he considered to differ by 30–50% on a daily basis. Beder (1995) examined eight approaches to computing VAR and found that the low and high estimates of VAR for a given portfolio differed by a factor that ranged from 6–14.
5. Marshall and Siegel asked a large number of VAR software vendors to estimate VAR for several portfolios. The distributional assumptions used in the estimations was presumably the same since all of the vendors used a common set of correlations and standard deviations provided by RiskMetrics.™ Despite common distributional assumptions, VAR estimates varied especially widely for nonlinear instruments suggesting that differences in the treatment of nonlinearities drive the differences in VAR estimates. For portfolios of foreign exchange options, which we will consider later, Marshall and Siegel found that the ratio of standard deviation of VAR estimate to median VAR estimate was 25%.
6. It is possible to combine Monte Carlo methods so that VAR with full repricing is applied to some sets of instruments, while other Monte Carlo techniques are applied to other options.
7. All of the information on the risk of the portfolio is contained in the function $G(.)$; a single VAR estimate uses only some of this information. However, as much of the function $G(.)$ as is desired can be recovered using VAR methods by computing VAR for whatever quantiles are desired. The accuracy of the methods will vary based on the quantile.

8. Changes in the value of an instrument are caused by changes in the factors and by instrument-specific idiosyncratic changes. In well-diversified portfolios only changes in the value of the factors should matter since idiosyncratic changes will be diversified away. If a portfolio is not well diversified, these idiosyncratic factors need to be treated as factors.
9. The general expression for the second term of the Taylor series is $\theta\Delta t$, where Δt is the time horizon for which VAR is computed. In our case this horizon has been normalized to one time period. Therefore, the second term in our first-order expansion appears as θ.
10. To illustrate the impact of factor choice on a first-order Taylor series ability to capture nonlinearity, consider a three-year bond with annual coupon c and principal \$1. Its price today is

$$B = cp(1) + cp(2) + (1 + c)p(3)$$

where the $p(i)$ are the respective prices of zero coupon bonds expiring i years from today. If the factors are the zero coupon bond prices, then the bond price is clearly a linear function of the factors, requiring at most a first-order Taylor series. If instead the factors are the one-, two- and three-year zero coupon interest rates z_1, z_2, and z_3 (RiskMetric™ provides information on zero coupon interest rates as factors, but not on zero coupon bond prices), then the expressions for δ and Γ are:

$$\delta = \begin{bmatrix} -ce^{(-z_1)} \\ -2ce^{(-2z_2)} \\ -3(1+c)e^{(-3z_3)} \end{bmatrix}$$

and

$$\Gamma = \begin{bmatrix} ce^{(-z_1)} & 0 & 0 \\ 0 & 4ce^{(-2z_2)} & 0 \\ 0 & 0 & 9(1+c)e^{-(3Z_3)} \end{bmatrix}$$

The example shows that the second set of factors requires a Taylor series of at least order 2 to capture the nonlinearity of changes in portfolio value.
11. Allen (1994) refers to the delta method as the correlation method; Wilson (1994) refers to this method as the delta-normal method.
12. The assumption that ε_{t+1} has mean 0 is appropriate over short time intervals such as a day or a week.
13. Anecdotal evidence suggests that many practitioners using delta-gamma methods set off-diagonal elements of Γ to zero in practice.
14. Changes in portfolio value in the delta method are approximated using linear combinations of normally distributed random variables. These are distributed normally. The delta-gamma method approximates changes in portfolio value using the sum of linear combinations of normally distributed random variables and second order quadratic terms. Because the quadratic terms have a chi-square distribution, the normality is lost.
15. This approach is one of those used in Jordan and Mackay (1995). They make their Monte Carlo draws using a historical simulation or bootstrap approach.
16. This distribution need not be normal.
17. If V is a function of two factors, then the shocks are ε_1, ε_2, ε_1^2, ε_2^2, and $\varepsilon_1\varepsilon_2$.
18. If two random variables are uncorrelated and normally distributed, then they are independent. ε_{t+1} and ε_{t+1}^2 are obviously not independent; thus they cannot be jointly normally distributed.
19. The computation of portfolio variance in the delta-gamma-delta approach is contained in the *RiskMetrics*™ *Technical Document*, 3rd edition, p. 137.
20. Here is one simple example where the assumption is violated. Suppose a firm has a long equity position. Then if one vector of factor shocks that lies outside the constraint set involves a large loss in the equity market, generating a large loss for the firm, then the opposite of this vector lies outside the constraint set too, but is likely to generate large increase in the value of the firm's portfolio. A positive proportion of the factor shocks outside the constraint set are near the one in the example, and will also have the same property.
21. The order of the Cornish–Fisher expansion determines the number of cumulants that are used.
22. This method is referred to as Structured Monte Carlo in *RiskMetrics*™ *Technical Document*, 3rd edition.

23. Allen (1994) describes a grid Monte Carlo approach used in combination with historical simulation. Estrella (1995) also discusses a grid approach.
24. Suppose H is a financial instrument whose value depends on the factor shocks ε_1, and ε_2, where H is highly nonlinear in ε_1 and nearly linear in ε_2. $\Delta H(\varepsilon_1, \varepsilon_2)$, can be approximated as a sum of a nonlinear function of ε_1 and a linear function of ε_2:

$$\begin{aligned}\Delta H(\varepsilon_1, \varepsilon_2) &= \Delta H(\varepsilon_1, 0) + [\Delta H(\varepsilon_1, \varepsilon_2) - \Delta H(\varepsilon_1, 0)] \\ &\approx \Delta H(\varepsilon_1, 0) + \Delta H_2(\varepsilon_1, 0)\varepsilon_2 \\ &\approx \Delta H(\varepsilon_1, 0) + \Delta H_2(0, 0)\varepsilon_2\end{aligned}$$

$$\text{where } \Delta H_2(\varepsilon_1, 0) = \frac{\partial}{\partial \varepsilon_2} \Delta H(\varepsilon_1, \varepsilon_2)|_{\varepsilon_2=0}$$

Line 2 in the above expression follows from a first-order Taylor series expansion of the term in braces in line 1. Line 3 follows from line 2 if the partial derivative of ΔH with respect to ε_2 is not very sensitive to the level of ε_1. This approximation also applies if ε_1 and ε_2 are vectors of changes in factors. However, the restriction that generates line 3 from line 2 has more bite when there are more factors.
25. If the factors are observable, then ε_t is observable, since

$$\varepsilon_t = f_t - f_{t-1}.$$

If ε_t is not observable, then the factors are not observable, and ε must be imputed some other way. This would invariably require some strong distributional assumptions that would undermine the advantages of using historical simulation in the first place.
26. It is important that the correct conditional distribution is used when making draws of ε, i.e., if today is considered a highly volatile period, then draws of ε should be made from a volatile historical period.
27. It is fortunate that confidence intervals can be constructed from a single set of N Monte Carlo draws. An alternative Monte Carlo procedure for constructing confidence intervals would involve computing standard errors for the Monte Carlo estimates by performing Monte Carlo simulations of Monte Carlo simulations, a very time-consuming process
28. The distinction made here between simple and complex computations is somewhat arbitrary. For example, when closed-form solutions are not available for instrument prices, pricing is probably more computationally intensive than performing the minimization computations required in the delta-gamma-minimization method. On the other hand, the minimization is more complicated than applying the Black–Scholes formula, but applying Black–Scholea is treated as complex while the minimization method is treated as simple.
29. If the portfolios being considered contain some instruments that are priced analytically while others are priced using 1 million Monte Carlo draws, the computational burdens repricing the two sets of instruments are very different and some allowance has to be made for this.
30. 100 identical options is one instrument, five options at different strike prices is five *different* instruments.
31. Each time a price is calculated, I assess one complex computation. Thus, I assume that $2n$ complex computations are required to compute δ because the numerical approximation for $\partial f(x)/\partial x = [f(x+h) - f(x-h)]/2h$, requires two repricings per factor; Similarly, I assume that $\partial^2 f(x)/\partial x^2 = [f(x+h) + f(x-h) - 2f(x)]/4h^2$ requires two repricings per factor and I assume that $\partial^2 f(x,y)/\partial x\, \partial y = [f(x+h, y+h) + f(x-h, y-h) - f(x-h, y+h) - f(x+h, y-h)]/4h^2$ requires four repricings per factor. For some instruments there are other methods of computing δ and Γ, this will change the number of complex computations.
32. The figures on numbers of complex computations for Γ presume that all of elements of the Γ matrix are computed. If some elements are restricted to be zero, this will substantially reduce the number of complex computations.
33. Although the number of complex computations required to compute Γ analytically is smaller than the number required to compute it numerically, the analytical expressions for the second derivative matrix may be so complicated that it is faster to compute Γ numerically than analytically.
34. The quadratic terms could be simplified using the transformation in Wilson (1994).

35. For example, in the regular grid Monte Carlo approach, for instruments that are priced using three factors, the grid contains 1000 points. In the modified grid Monte Carlo approach, the grid will always contain ten points.
36. The relative number of complex computations required using the different approaches will vary for different portfolios; i.e., for some portfolios grid Monte Carlo will require more complex computations while for others full Monte Carlo will require more complex computations.
37. A potential problem with this parallel approach is that it removes an element of independence from the risk management function, and raises the question of how much independence is necessary and at what cost?
38. Similarly, it is possible to generate Monte Carlo draws of the factor shocks, and then for each Monte Carlo draw approximate ΔV for the portfolio by adding together estimates of the change in portfolio value for different parts of the portfolio. For example, a second-order Taylor expansion can be used to approximate the change in the value of some instruments, while full repricing can be used with other instruments.
39. It is important to emphasize that the gamma matrix used in the VAR methods below is the *matrix* of second and cross-partials of the change in portfolio value due to a change in the factors.
40. These additional results are available from the author upon request.
41. The error of VAR estimate $V\hat{A}R$ is $V\hat{A}R - VAR$, and its percentage error is $\frac{V\hat{A}R - VAR}{VAR}$.
42. True overstatement is not known since VAR is measured with error.
43. Suppose one VAR method makes small and identical mistakes for 99 portfolios, and makes a large mistake for one portfolio, while for the same portfolios the ''other'' method makes no mistakes for 98 portfolios, and makes one small mistake and one large mistake. Clearly the ''other'' method produces strictly superior VAR estimates. However, conditional on making a mistake, the ''other'' method has a larger mean and variance than the method which is clearly inferior.
44. Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) can be computed from mean, standard deviation, and frequency of under- and overstatement. Errors that are statistically indistinguishable from 0 are treated as 0 and not included. The formulas for the loss functions are:

$$\mathrm{MAE} = \mathrm{MU} * \mathrm{FREQU} + \mathrm{MO} * \mathrm{FREQO},$$
$$\mathrm{RMSE} = \sqrt{(\mathrm{MU}^2 + \mathrm{STDU}^2) * \mathrm{FREQU} + (\mathrm{MO}^2 + \mathrm{STDO}^2) * \mathrm{FREQO}}$$

where MU, STDU, and FREQU are mean, standard deviation, and frequency of understatement, respectively, and MO, STDO, and FREQO are mean, standard deviation, and frequency of overstatement, respectively.
45. There are no three-dimensional figures for simulation exercise 2 because the portfolios vary in too many dimensions.
46. For example, to price change in portfolio value from a grid for each option, a shock's location on the grid needed to be computed for each Monte Carlo draw and each option. This added considerable time to the grid Monte Carlo method.
47. The assumption that the Monte Carlo lower bound for Value at Risk is greater than 0 is important. The assumption is reasonable for most risky portfolios. If it is not true for a given number of Monte Carlo draws, but true Value at Risk is believed to be greater than zero, then additional Monte Carlo draws should eventually produce a lower bound that is greater than 0. If it is still not possible to produce a lower bound that is greater than 0, then it will not be possible to create meaningful bounds for the percentage errors, but it will be possible to create bounds for the actual errors.
48. For options on some currency pairs RiskMetrics™ does not provide volatility information. For example, the variance of the French franc/deutschemark (FRF/DM) exchange rate is not provided. To calculate this standard deviation from the data provided by RiskMetrics, let $F = \$/\mathrm{FFR}$, $\mathrm{O} = \$/\mathrm{DM}$, and let $G = \mathrm{FFR}/\mathrm{DM}$. It follows that:

$$G = D/F$$
$$\ln(G) = \ln(D) - \ln(F)$$
$$dG/G = dD/D - dF/F$$
$$\mathrm{Var}(dG/G) = \mathrm{Var}(dD/D) + \mathrm{Var}(dF/F) - 2 * \mathrm{Cov}(dD/D, dF/F)$$

VAR (dG/G) can be computed in RiskMetrics™ since all the expressions on the right-hand side of the last expression can be computed using RiskMetrics.™

49. U and V are chosen so that

$$\text{Prob}(U \geq \hat{VAR} - VAR \geq L) \geq 0.95,$$

and $U\%$ and $L\%$ are chosen so that

$$\text{Prob}\left(U\% \geq \frac{\hat{V}AR - VAR}{VAR} \geq L\%\right) \geq 0.95.$$

50. The amount of understatement and percentage understatement are at least as big as U and U% respectively with probability exceeding 95%.

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